

THE METHOD 1.6 — THE INDEX OF INDICES

Generated on 2026-08-15 by `indices.py`. Every index this work builds on or beside the atomic index — what it holds, whether it closes, whether it carries time, and what role it plays for Λ . Measured where measurable; where an index is named and never built, that is stated rather than filled in.

I • THE ATOMIC INDEX ITSELF

$\Lambda = \{ (n, \ell, k, q, e, f, g, 2S) \in \mathbb{Z}^8 : \text{eight constraints, at caps} \}$.

Λ_8 · eight coordinates · seven constraints · a caterpillar with 2S pendant at k

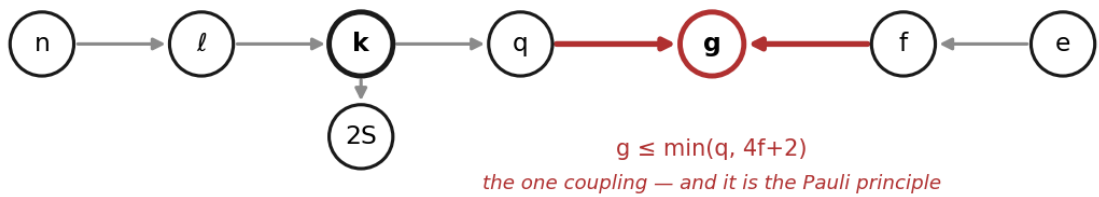


Figure 1. Λ_8 's constraint graph. Seven bounds on eight coordinates — a caterpillar with 2S pendant at k. Every edge is a single monotone inequality except one: **g takes two parents**, and $g \leq \min(q, 4f+2)$ is the only non-product term in the whole expression.

A cell is a **transition**: a source subshell (n, ℓ) holding k electrons, q of them moving, into a target subshell (e, f) that ends with g, at spin 2S. **Not a state — a move.** Everything else in this document follows from that.

constraint	reads
$\ell \leq n - 1$	hydrogenic — the subshell fits its shell
$k \leq 4\ell + 2$	Pauli — the subshell's capacity
$q \leq k$	conservation of the removed — you cannot take more than is held
$f \leq e - 1$	hydrogenic again, on the target
$g \leq 4f + 2$	Pauli again, on the target
$g \leq q$	conservation of the placed
$2S \leq k$	the multiplicity a subshell can carry
$k \geq 1$	*definitional* — a transition needs a mover

The one coupling is $g \leq \min(q, 4f+2)$, the single non-product term in the generating function, and it is the Pauli principle. Remove it and 976 becomes 1,000; the 24 excluded cells all have $f = 0$ and $g = 3$ — three electrons in an s orbital.

II • THE TOWER

Each stage adds one axis and inherits every bound below.

stage	cells	axes	composable	fraction	the axis added
Λ_8	976	8	0	0.0000	—
Λ_9	1,654	9	1,169	0.7068	$2S' \leq g$
Λ_{10}	2,535	10	2,050	0.8087	$2S' \leq v \leq g$
Λ_{11}	13,585	11	9,450	0.6956	$2J_c \leq \hat{\varphi}(k)$
Λ_{12}	22,275	12	14,242	0.6394	$2K \leq 2J_c + 2f$

stage	cells	axes	composable	fraction	the axis added
Λ_{13}	64,290	13	39,772	0.6186	\

2J – 2K\

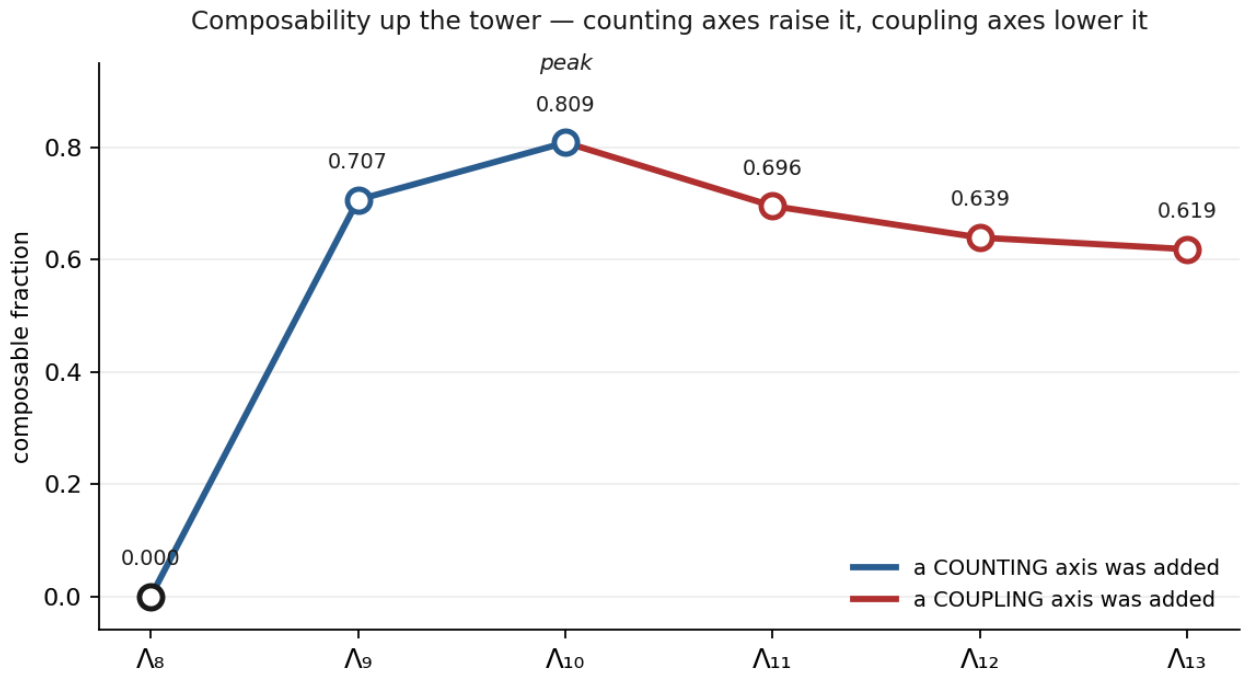


Figure 2. The composable fraction stage by stage. Λ_8 composes not at all — four source coordinates against three target. The two counting axes take it to a peak of 0.8087 at Λ_{10} ; the three coupling axes lower it every time.

Counting axes raise the composable fraction; coupling axes lower it. No exception.

Λ_8 composes not at all — four source coordinates against three target. Λ_{10} is the peak at 80.87%, and every stage after it falls. The non-composable cells are exactly those with $g = 0$ at the peak, since $k \geq 1$ forbids an empty target being a source.

III • THE INDEXES DRAWN BESIDE IT

index	coordinates	cells	box	E	kind
the periodic table	(period, group)	90	126	36	state
Janet's left-step	($n+l$, Z)	118	944	0	state
the calendar	(month, day)	365	372	7	state
a box ordering	(l, w, h)	35	125	0	state
a chessboard	(rank, file)	64	64	0	state

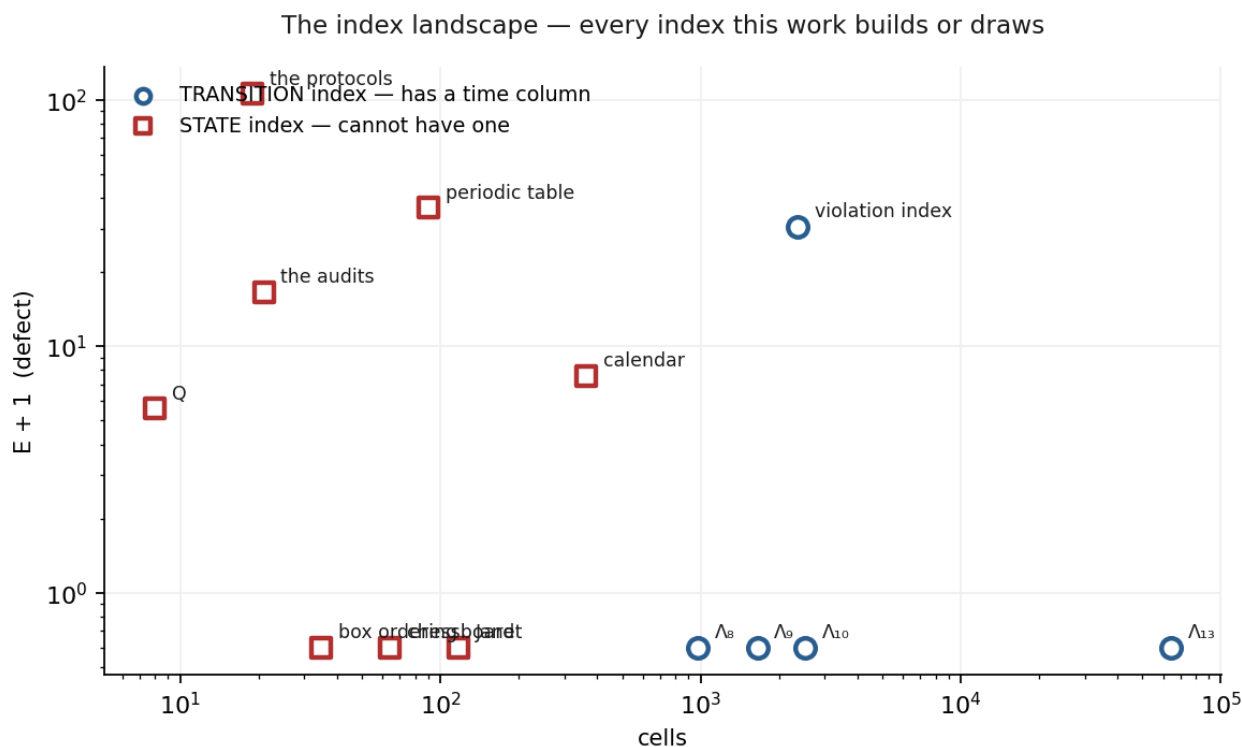


Figure 3. Every index this work builds or draws, by cells against defect. **Circles are transition indexes** — their cells are moves, so they carry a time column. **Squares are state indexes** — a cell is one position, and the question of composition does not arise.

None of these is a transition index, so none can carry a time column. A state cell has one position; the question *is my target another cell's source* does not arise.

Janet closes at $E = 0$ and the classroom table does not. That is a fact about the two tables **in coordinates chemists fixed for other reasons** — §21.6 shows any composite count has SOME coordinate system where $E = 0$, so the result is that Janet's 1928 ordering, chosen for shell filling, happens to close.

IV • THE VIOLATION INDEX

The companion paper's object, indexed over nine letters. It is here because it is the one index this work reasons about without holding — **its cells are not printed anywhere**, and what follows is everything that is.

letter	meaning	rungs
X	exotic matter required	4
S	semiclassical corrections	3
IC	initial conditions	3
U	unitarity	3
NEC	null energy condition points	5
L	locality	2
SD	superdeterminism	2
DNc	dynamical, continuous	3
DNd	dynamical, discrete	3

$4 \cdot 3 \cdot 3 \cdot 3 \cdot 5 \cdot 2 \cdot 2 \cdot 3 \cdot 3 = 19,440$, and the index holds **2,370** — 12.19% density.

What the companion prints

object	threshold	cells	reachable
classical black hole	none	2,370	1,410
Hawking-evaporating	$NEC \geq 1$	2,196	1,410
Planck-scale wormhole	$NEC \geq 2$	1,764	1,134
macroscopic wormhole	$NEC \geq 3$	1,146	738
universal horizon	$X \geq 2$	1,374	840
time machine	$X = 3$	558	360

Twelve numbers, and one column is a second predicate on the same cells. Differenced, the cells give $NEC = 0$ at 174, $NEC = 1$ at 432, $NEC = 2$ at 618, $NEC \geq 3$ at 1,146. **All 174 cells at $NEC = 0$ are unreachable** — removing them removes no reachable cell — and **exactly 64% of $NEC = 1$ and 64% of $NEC = 2$ cells are reachable**, 276 of 432 and 396 of 618.

Its defect, and the one constraint printed

$E = 30$, collapsing as 1×30 — one core cell repeated across the free coordinates — with the core at **$(X = 0, U = 0, NEC = 3)$** . At fifteen letters: 18,072 cells, **$E = 816 = 1 \times 816$** . **Multiplicities 3, 5, 10, 30 at 4, 5, 7 and 9 coordinates**, and 5 is not a product of any two rung counts, so **the edge list constrains the free letters as well as the support**.

The constraint the companion prints, at arity 3:

$$NEC \geq 3 \wedge X = 0 \rightarrow U \geq 1 \text{ — cells } 2,370, E = 30, \text{ core } 1$$

with three variants: $NEC \geq 3 \rightarrow U \geq 1$ at arity 2 giving **$E = 0$** , and $NEC \geq 3 \wedge U = 0 \rightarrow X \geq 1$ and $NEC \geq 3 \rightarrow (IC \vee U \vee X)$ both giving **$E = 30$** . **A jurisdicted forcing and an unjurisdicted disjunction give identical defects; arity 2 gives none.**

What is not printed, and what nine searches established

The edge list itself. §5.7 of the companion says so outright. Nine searches were run against the twelve numbers; the best holds **cells exactly 2,370 at distance 29**, with the whole residual on the NEC interior — **$NEC \geq 2$ sixteen too high, $NEC \geq 3$ nine too low, missing in opposite directions.**

Eight alphabets were tried and none moved it: simple implications, disjunctive heads, conjunctive bodies, all 47 single-rule edits, four alternative rung readings, two threshold semantics, seeding at $NEC = 3$, and weighted sums with gated deviations.

The rungs and the $\geq$ reading are confirmed correct — every alternative is two orders of magnitude worse.

And the reachable column reduces to the same missing artefact. Three local readings of reachability give ~99% against a printed 60%; the companion's own *here?* column says reachable means reachable FROM HERE, directed from one position in a transition graph. **Both columns need the edge list.**

V • THE INDEXES BUILT FROM IT

Each of these exists because Λ exists — it indexes something Λ 's cells have, or something the act of building Λ produced.

The electromagnetic quotient

Λ modulo the dipole selection rules — $\Delta\ell = \pm 1$, $\Delta S = 0$, parity. A quotient, not a subset: cells are identified when the rules cannot distinguish them.

Its $E = 0$ is VACUOUS — §12.11.8 — because the image is a complete rectangle. **It closes because it is a box, not because the selection rules constrain it**, and that is the one closure in this work that certifies nothing.

The crossing — allowed transitions compose least within an atom and most between atoms

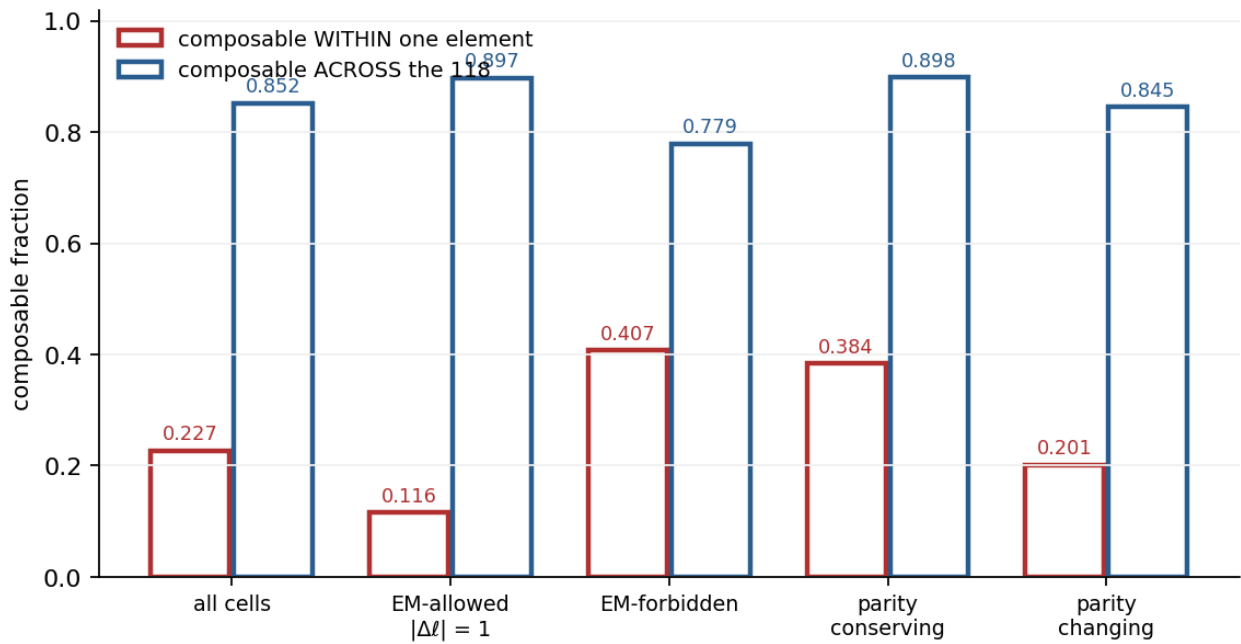


Figure 4. The crossing. Within one atom a dipole-allowed transition is the LEAST likely to be followable; across the 118 elements it is the most. The rule that forbids composition inside an atom is the rule that enables it between atoms.

Its measured role is the crossing. Within one element, EM-allowed transitions compose at **11.6%** against the forbidden **40.7%**. Across the 118 elements the order reverses — **89.7% against 77.9%**. The rule that forbids composition inside an atom is the rule that enables it between atoms.

The time index

Not a coordinate. The second column. §12.11.0: composition IS the temporal order — a cell's source end is a before, its target end an after, and q is what changed between them.

An index has a time column exactly when its cells are moves. Λ_8 976/0, Λ_9 1,654/1,169, Λ_{10} 2,535/2,050 — and the periodic table and the calendar **cannot have one**, because a state cell has one position and there is nothing to compose.

What Λ lacks is not a clock but a denominator — claims per session — which §12.11.1.1 now supplies. A date must never enter Λ : a transition is a type, and types are not dated.

The space index

The first column. What the index holds, as against what it can compose. Λ has no spatial coordinate: n and e are shell numbers, not positions, and nothing in the eight is a place.

The nearest thing to a spatial reading is the observability one: an observer's register is Λ -valued because observation IS an atomic state change, so an unregistered event is an admitted-and-absent cell and **E(local register) measures what is not yet real to that observer.** That is a definition with a falsifiable consequence and it is not yet built.

The languages

A language is a coordinate system; translation is re-coordinatisation; E is the cost.

language	closure operator	delete	add
order	$\mathcal{R}$	restored	absorbed
geometry	monotone polyhedron	restored	absorbed
algebra / logic	ideal, Gröbner basis	still derivable	T-h enters the ideal
analysis	coefficients of F	restored	absorbed

language	closure operator	delete	add
information	description length	E_bits stays 0	grows
statistics	max-entropy on the marginals	restored	grows
documentary	none — no algorithm	—	—

Six agree on Λ at 976 and all ten pairs hold — $C(5,2)$, a complete graph, not a ladder. **Statistics was tested and qualifies with a caveat:** it recovers Λ at 976 and $E = 0$, but gives **$E = 0$ on the periodic table where $\mathcal{R}$ gives 36** — *marginals cannot see a hole*. It agrees with the others only where there is nothing to disagree about.

The book's own indexes

index	coordinates	cells	E	what it holds
the audits	object<source<artefact<outside · what · how · depends	21	16	twenty-two audits
the register	corroboration $\leq \varphi(\text{repair})$	33	6	the corrections
Q	blocks · obstacle · cost · depends	8	5	what the book does not know
the protocols	trigger · object · failure · earned	19	105	twenty-four protocols
the constraints	parents · form · carried · source · role	15	21 / E₄ 6	every constraint in the book
the mathematics	kind · language · status · verification · precedent	27	0	fibred, sixteen fibres
the numbers	fibre · kind · role	—	40	every figure the book prints
the channel survey	$Z \cdot \text{core charge} \cdot \ell$	1,664	1,351	596 Rydberg channels across 313 cells

These exist because the book exists, not because Λ does — but they are built by Λ 's method, and the audits index's two frontier cells are the same pair $\mathcal{R}_4$ refuses at §14.5.6, reached by two computations sharing no code.

How much of an index determines the rest

Remove cells at random and ask whether $\mathcal{R}$ puts them back. The largest fraction removable with exact recovery is a property of the index, and it had never been measured for any of these.

index	dimension	envelopes	coupling	redundancy
Λ	8	56	29%	61%
$\Lambda_{\text{spectra}}^{\text{obs}}$	3	6	17%	20%
a box ordering	3	6	50%	0%
Janet	2	2	50%	0%
the periodic table	2	2	0%	0%
the calendar	2	2	0%	0%

Coupling does not explain it. The fraction of coordinate pairs whose envelope actually constrains gives **$r^2 = 0.002$, $p = 0.94$** against redundancy: Janet and the box ordering both couple at 50% and recover nothing, while $\Lambda_{\text{spectra}}^{\text{obs}}$ couples at 17% and recovers a fifth. *That conjecture was stated before the measurement and is wrong.*

DIMENSION explains it, and the projection test shows so on a single object. Λ restricted to its own first d coordinates, constraints unchanged:

$d = 8$ 61% $d = 5$ 30% $d = 7$ 30% $d = 4$ 5% $d = 6$ 30% $d = 3$ 0%

$\mathcal{R}$ works on **pairwise** envelopes, so an index of dimension d carries $d(d-1)$ of them — **56 for Λ , 6 for $\Lambda_{\text{spectra}}^{\text{obs}}$, 2 for a plane**. With two envelopes there is almost nothing to reconstruct from, and that is what the operator is rather than a fact about any subject.

But only INDEPENDENT coordinates count. Adding the electron count $N_e = Z - c + 1$ to $\Lambda_{\text{spectra}}^{\text{obs}}$ — a function of two coordinates it already has — takes redundancy from **20% to 0%**, while doubling the envelopes and raising coupling from 17% to 33%. *A derived coordinate adds no information and obliges $\mathcal{R}$ to reproduce it exactly, so recovery becomes strictly harder.* Registers 1119–1122.

The channel survey — $\Lambda_{\text{spectra}}^{\text{obs}}$

Coordinates: $Z \cdot \text{core charge} \cdot \ell$. Each cell is a Rydberg channel — a fixed parent core, a fixed ℓ , n running. The compendium holds **596 channels across 313 of the survey's 1,664 cells**.

This is not the coordinate index, and the two were once both called Λ_{spectra} . Λ_{spectra} proper is the four-coordinate index (Z , charge, ℓ , $2S+1$) at 104,832 cells with $E = 0$ — see Λ_{spectra} — *the channel index**, below. $\Lambda_{\text{spectra}}^{\text{obs}}$ is the *three-coordinate survey of what has been measured*. The column beside each cap below is grid minus held — the **unwitnessed count** — and not* a closure defect: reading it as E would say the index is open when it is a fixed point. R 1657.

Where its alphabets come from, and why that differs from Λ . Λ 's coordinates are bounded by the physics — $\ell < n$, $k \leq 4\ell+2$ — and all 118 ground configurations are known within its caps, so Λ is complete at 976 cells. **$\Lambda_{\text{spectra}}^{\text{obs}}$ is not.** Rydberg series are unbounded in n and ℓ runs to $n-1$, so the index is **infinite unless capped**, and every cap is a decision. Three defensible ones give:

alphabet	grid	unwitnessed
as measured — charge [1, 2, 3, 4, 5, 6, 9, 11, 15, 16]	1,664	1,351
charge 1 to $Z-1$ over the elements held	4,376	4,148
all 118 elements, ℓ 0–7	55,224	54,996

All three grids are closed — each is a fixed point of $\mathcal{R}$, verified by double projection. The number beside each is what has *not* been measured under that cap, so it is a result about the cap and about the state of the literature; it is never a closure defect*. Registers 959–962, 1657.*

What the index says about cells it does not hold. Every cell carries a grade, and the grade records how far the statement travelled:

grade	cells	what it means
MEASURED	205	a defect computed from levels
BRACKETED	5	bounded above and below by DIFFERENT mechanisms
BOUNDED	42	bounded on one side, adjacent to a measurement
PROPAGATED	954	a bound inherited through a chain of 2 to 13 steps
FORMAL	73	the only bound is one a mechanism already implies
UNCONSTRAINED	33	no statement at all

97.5% of the index carries a defensible statement. The three relations that propagate are **P.lcollapse** along ℓ , **P.iso** along an isoelectronic sequence, and **P.charge** along the charge states of one element. Registers 963–965, 981.

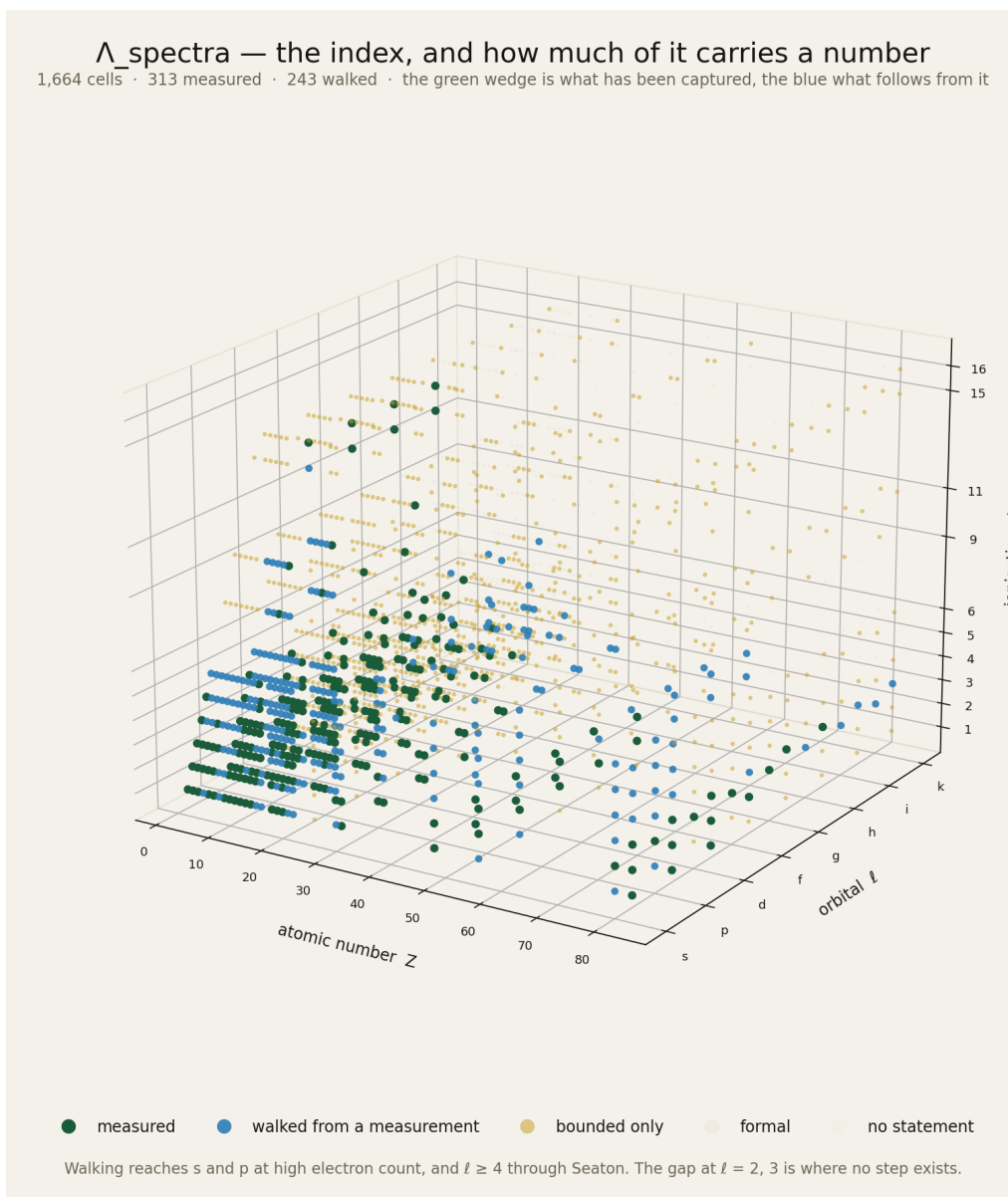


Figure 5. Λ _{spectra}^{obs} drawn as a lattice — element across, ℓ into the page, ionisation stage up. Green is measured, orange bounded from two sides by different mechanisms, gold from one; the faint markers are cells the structure admits and nobody has measured. **The occupied region is a wedge at low Z and low ℓ** , which is where long Rydberg series have been tabulated — and the emptiness above it is the shape of what an index says it does not know.

And the last 33 are of three kinds, only two of which a capture can fix. P I, Cd I, Ti II and Ar I want longer captures. **Sc I and Ti I have no Rydberg series at all** — their levels are $3d.4s.4p$ and $3d^2.4s$ mixtures with no n running, so fourteen cells are unreachable by any capture. *Register* 983. **The ceiling is 98.6%, not 100%, and that is a fact about the open-shell transition metals rather than about the collection.**

VI · THE INDEXES NAMED AND NOT BUILT

§29.1: the absence is not evidence. These were named in the course of the work and no index exists for them. Saying so is the point of this section.

Sound. Phonon modes are discrete and indexable — ω , lattice momentum, polarisation, occupation — but they index a **material**, not an atom. **No such index is built here**, and it would sit beside Λ rather than inside it, since Λ has no coordinate a lattice vibration could occupy.

Frequency. A transition's frequency is $\Delta E/h$, and ΔE is not a coordinate of Λ — §6.2 gives first ionisation energies to the bracket and the bracket REFUSES, at Be→B, N→O, Mg→Al and P→S, four steps of twelve. *That refusal is the book's most direct statement about frequency: the coordinates cannot bracket it, and where they refuse a mechanism is entering.*

The multiverse. §32.6.1 places this outside all four of the book's indexes, with the destination coordinate, because §E.1.4 shows **the open set cannot express an unclosable question.** *It is not an index that has not been built. It is the shape of a question this work cannot pose* — and Theorem 12.1 is why: Λ 's order structure contributes nothing predictively beyond what the coordinates supply, and its coordinates are n, ℓ , k, q, e, f, g, 2S.

VII • THE FULL TABLES

Every index small enough to print, printed. The tower above Λ_8 runs to 64,290 cells at Λ_{13} and is given by its construction instead; the violation index has no cells to give.

Λ_8 — all 976 cells

#	n	ℓ	k	q	e	f	g	2S
1	1	0	1	0	1	0	0	0
2	1	0	1	0	1	0	0	1
3	1	0	1	0	2	0	0	0
4	1	0	1	0	2	0	0	1
5	1	0	1	0	2	1	0	0
6	1	0	1	0	2	1	0	1
7	1	0	1	0	3	0	0	0
8	1	0	1	0	3	0	0	1
9	1	0	1	0	3	1	0	0
10	1	0	1	0	3	1	0	1
11	1	0	1	1	1	0	0	0
12	1	0	1	1	1	0	0	1
13	1	0	1	1	1	0	1	0
14	1	0	1	1	1	0	1	1
15	1	0	1	1	2	0	0	0
16	1	0	1	1	2	0	0	1
17	1	0	1	1	2	0	1	0
18	1	0	1	1	2	0	1	1
19	1	0	1	1	2	1	0	0
20	1	0	1	1	2	1	0	1
21	1	0	1	1	2	1	1	0
22	1	0	1	1	2	1	1	1
23	1	0	1	1	3	0	0	0
24	1	0	1	1	3	0	0	1
25	1	0	1	1	3	0	1	0
26	1	0	1	1	3	0	1	1
27	1	0	1	1	3	1	0	0
28	1	0	1	1	3	1	0	1
29	1	0	1	1	3	1	1	0
30	1	0	1	1	3	1	1	1
#	n	ℓ	k	q	e	f	g	2S
31	1	0	2	0	1	0	0	0

#	n	ℓ	k	q	e	f	g	2S
32	1	0	2	0	1	0	0	1
33	1	0	2	0	1	0	0	2
34	1	0	2	0	2	0	0	0
35	1	0	2	0	2	0	0	1
36	1	0	2	0	2	0	0	2
37	1	0	2	0	2	1	0	0
38	1	0	2	0	2	1	0	1
39	1	0	2	0	2	1	0	2
40	1	0	2	0	3	0	0	0
41	1	0	2	0	3	0	0	1
42	1	0	2	0	3	0	0	2
43	1	0	2	0	3	1	0	0
44	1	0	2	0	3	1	0	1
45	1	0	2	0	3	1	0	2
46	1	0	2	1	1	0	0	0
47	1	0	2	1	1	0	0	1
48	1	0	2	1	1	0	0	2
49	1	0	2	1	1	0	1	0
50	1	0	2	1	1	0	1	1
51	1	0	2	1	1	0	1	2
52	1	0	2	1	2	0	0	0
53	1	0	2	1	2	0	0	1
54	1	0	2	1	2	0	0	2
55	1	0	2	1	2	0	1	0
56	1	0	2	1	2	0	1	1
57	1	0	2	1	2	0	1	2
58	1	0	2	1	2	1	0	0
59	1	0	2	1	2	1	0	1
60	1	0	2	1	2	1	0	2

#	n	ℓ	k	q	e	f	g	2S
61	1	0	2	1	2	1	1	0
62	1	0	2	1	2	1	1	1
63	1	0	2	1	2	1	1	2
64	1	0	2	1	3	0	0	0
65	1	0	2	1	3	0	0	1
66	1	0	2	1	3	0	0	2
67	1	0	2	1	3	0	1	0
68	1	0	2	1	3	0	1	1
69	1	0	2	1	3	0	1	2
70	1	0	2	1	3	1	0	0
71	1	0	2	1	3	1	0	1
72	1	0	2	1	3	1	0	2
73	1	0	2	1	3	1	1	0
74	1	0	2	1	3	1	1	1
75	1	0	2	1	3	1	1	2
76	1	0	2	2	1	0	0	0
77	1	0	2	2	1	0	0	1

#	n	ℓ	k	q	e	f	g	2S
78	1	0	2	2	1	0	0	2
79	1	0	2	2	1	0	1	0
80	1	0	2	2	1	0	1	1
81	1	0	2	2	1	0	1	2
82	1	0	2	2	1	0	2	0
83	1	0	2	2	1	0	2	1
84	1	0	2	2	1	0	2	2
85	1	0	2	2	2	0	0	0
86	1	0	2	2	2	0	0	1
87	1	0	2	2	2	0	0	2
88	1	0	2	2	2	0	1	0
89	1	0	2	2	2	0	1	1
90	1	0	2	2	2	0	1	2

#	n	ℓ	k	q	e	f	g	2S
91	1	0	2	2	2	0	2	0
92	1	0	2	2	2	0	2	1
93	1	0	2	2	2	0	2	2
94	1	0	2	2	2	1	0	0
95	1	0	2	2	2	1	0	1
96	1	0	2	2	2	1	0	2
97	1	0	2	2	2	1	1	0
98	1	0	2	2	2	1	1	1
99	1	0	2	2	2	1	1	2
100	1	0	2	2	2	1	2	0
101	1	0	2	2	2	1	2	1
102	1	0	2	2	2	1	2	2
103	1	0	2	2	3	0	0	0
104	1	0	2	2	3	0	0	1
105	1	0	2	2	3	0	0	2
106	1	0	2	2	3	0	1	0
107	1	0	2	2	3	0	1	1
108	1	0	2	2	3	0	1	2
109	1	0	2	2	3	0	2	0
110	1	0	2	2	3	0	2	1
111	1	0	2	2	3	0	2	2
112	1	0	2	2	3	1	0	0
113	1	0	2	2	3	1	0	1
114	1	0	2	2	3	1	0	2
115	1	0	2	2	3	1	1	0
116	1	0	2	2	3	1	1	1
117	1	0	2	2	3	1	1	2
118	1	0	2	2	3	1	2	0
119	1	0	2	2	3	1	2	1
120	1	0	2	2	3	1	2	2

#	n	ℓ	k	q	e	f	g	2S
121	2	0	1	0	1	0	0	0
122	2	0	1	0	1	0	0	1

#	n	ℓ	k	q	e	f	g	2S
123	2	0	1	0	2	0	0	0
124	2	0	1	0	2	0	0	1
125	2	0	1	0	2	1	0	0
126	2	0	1	0	2	1	0	1
127	2	0	1	0	3	0	0	0
128	2	0	1	0	3	0	0	1
129	2	0	1	0	3	1	0	0
130	2	0	1	0	3	1	0	1
131	2	0	1	1	1	0	0	0
132	2	0	1	1	1	0	0	1
133	2	0	1	1	1	0	1	0
134	2	0	1	1	1	0	1	1
135	2	0	1	1	2	0	0	0
136	2	0	1	1	2	0	0	1
137	2	0	1	1	2	0	1	0
138	2	0	1	1	2	0	1	1
139	2	0	1	1	2	1	0	0
140	2	0	1	1	2	1	0	1
141	2	0	1	1	2	1	1	0
142	2	0	1	1	2	1	1	1
143	2	0	1	1	3	0	0	0
144	2	0	1	1	3	0	0	1
145	2	0	1	1	3	0	1	0
146	2	0	1	1	3	0	1	1
147	2	0	1	1	3	1	0	0
148	2	0	1	1	3	1	0	1
149	2	0	1	1	3	1	1	0
150	2	0	1	1	3	1	1	1

#	n	ℓ	k	q	e	f	g	2S
151	2	0	2	0	1	0	0	0
152	2	0	2	0	1	0	0	1
153	2	0	2	0	1	0	0	2
154	2	0	2	0	2	0	0	0
155	2	0	2	0	2	0	0	1
156	2	0	2	0	2	0	0	2
157	2	0	2	0	2	1	0	0
158	2	0	2	0	2	1	0	1
159	2	0	2	0	2	1	0	2
160	2	0	2	0	3	0	0	0
161	2	0	2	0	3	0	0	1
162	2	0	2	0	3	0	0	2
163	2	0	2	0	3	1	0	0
164	2	0	2	0	3	1	0	1
165	2	0	2	0	3	1	0	2
166	2	0	2	1	1	0	0	0
167	2	0	2	1	1	0	0	1
168	2	0	2	1	1	0	0	2

#	n	ℓ	k	q	e	f	g	2S
169	2	0	2	1	1	0	1	0
170	2	0	2	1	1	0	1	1
171	2	0	2	1	1	0	1	2
172	2	0	2	1	2	0	0	0
173	2	0	2	1	2	0	0	1
174	2	0	2	1	2	0	0	2
175	2	0	2	1	2	0	1	0
176	2	0	2	1	2	0	1	1
177	2	0	2	1	2	0	1	2
178	2	0	2	1	2	1	0	0
179	2	0	2	1	2	1	0	1
180	2	0	2	1	2	1	0	2

#	n	ℓ	k	q	e	f	g	2S
181	2	0	2	1	2	1	1	0
182	2	0	2	1	2	1	1	1
183	2	0	2	1	2	1	1	2
184	2	0	2	1	3	0	0	0
185	2	0	2	1	3	0	0	1
186	2	0	2	1	3	0	0	2
187	2	0	2	1	3	0	1	0
188	2	0	2	1	3	0	1	1
189	2	0	2	1	3	0	1	2
190	2	0	2	1	3	1	0	0
191	2	0	2	1	3	1	0	1
192	2	0	2	1	3	1	0	2
193	2	0	2	1	3	1	1	0
194	2	0	2	1	3	1	1	1
195	2	0	2	1	3	1	1	2
196	2	0	2	2	1	0	0	0
197	2	0	2	2	1	0	0	1
198	2	0	2	2	1	0	0	2
199	2	0	2	2	1	0	1	0
200	2	0	2	2	1	0	1	1
201	2	0	2	2	1	0	1	2
202	2	0	2	2	1	0	2	0
203	2	0	2	2	1	0	2	1
204	2	0	2	2	1	0	2	2
205	2	0	2	2	2	0	0	0
206	2	0	2	2	2	0	0	1
207	2	0	2	2	2	0	0	2
208	2	0	2	2	2	0	1	0
209	2	0	2	2	2	0	1	1
210	2	0	2	2	2	0	1	2

#	n	ℓ	k	q	e	f	g	2S
211	2	0	2	2	2	0	2	0
212	2	0	2	2	2	0	2	1
213	2	0	2	2	2	0	2	2

#	n	ℓ	k	q	e	f	g	2S
214	2	0	2	2	2	1	0	0
215	2	0	2	2	2	1	0	1
216	2	0	2	2	2	1	0	2
217	2	0	2	2	2	1	1	0
218	2	0	2	2	2	1	1	1
219	2	0	2	2	2	1	1	2
220	2	0	2	2	2	1	2	0
221	2	0	2	2	2	1	2	1
222	2	0	2	2	2	1	2	2
223	2	0	2	2	3	0	0	0
224	2	0	2	2	3	0	0	1
225	2	0	2	2	3	0	0	2
226	2	0	2	2	3	0	1	0
227	2	0	2	2	3	0	1	1
228	2	0	2	2	3	0	1	2
229	2	0	2	2	3	0	2	0
230	2	0	2	2	3	0	2	1
231	2	0	2	2	3	0	2	2
232	2	0	2	2	3	1	0	0
233	2	0	2	2	3	1	0	1
234	2	0	2	2	3	1	0	2
235	2	0	2	2	3	1	1	0
236	2	0	2	2	3	1	1	1
237	2	0	2	2	3	1	1	2
238	2	0	2	2	3	1	2	0
239	2	0	2	2	3	1	2	1
240	2	0	2	2	3	1	2	2

#	n	ℓ	k	q	e	f	g	2S
241	2	1	1	0	1	0	0	0
242	2	1	1	0	1	0	0	1
243	2	1	1	0	2	0	0	0
244	2	1	1	0	2	0	0	1
245	2	1	1	0	2	1	0	0
246	2	1	1	0	2	1	0	1
247	2	1	1	0	3	0	0	0
248	2	1	1	0	3	0	0	1
249	2	1	1	0	3	1	0	0
250	2	1	1	0	3	1	0	1
251	2	1	1	1	1	0	0	0
252	2	1	1	1	1	0	0	1
253	2	1	1	1	1	0	1	0
254	2	1	1	1	1	0	1	1
255	2	1	1	1	2	0	0	0
256	2	1	1	1	2	0	0	1
257	2	1	1	1	2	0	1	0
258	2	1	1	1	2	0	1	1
259	2	1	1	1	2	1	0	0

#	n	ℓ	k	q	e	f	g	2S
260	2	1	1	1	2	1	0	1
261	2	1	1	1	2	1	1	0
262	2	1	1	1	2	1	1	1
263	2	1	1	1	3	0	0	0
264	2	1	1	1	3	0	0	1
265	2	1	1	1	3	0	1	0
266	2	1	1	1	3	0	1	1
267	2	1	1	1	3	1	0	0
268	2	1	1	1	3	1	0	1
269	2	1	1	1	3	1	1	0
270	2	1	1	1	3	1	1	1

#	n	ℓ	k	q	e	f	g	2S
271	2	1	2	0	1	0	0	0
272	2	1	2	0	1	0	0	1
273	2	1	2	0	1	0	0	2
274	2	1	2	0	2	0	0	0
275	2	1	2	0	2	0	0	1
276	2	1	2	0	2	0	0	2
277	2	1	2	0	2	1	0	0
278	2	1	2	0	2	1	0	1
279	2	1	2	0	2	1	0	2
280	2	1	2	0	3	0	0	0
281	2	1	2	0	3	0	0	1
282	2	1	2	0	3	0	0	2
283	2	1	2	0	3	1	0	0
284	2	1	2	0	3	1	0	1
285	2	1	2	0	3	1	0	2
286	2	1	2	1	1	0	0	0
287	2	1	2	1	1	0	0	1
288	2	1	2	1	1	0	0	2
289	2	1	2	1	1	0	1	0
290	2	1	2	1	1	0	1	1
291	2	1	2	1	1	0	1	2
292	2	1	2	1	2	0	0	0
293	2	1	2	1	2	0	0	1
294	2	1	2	1	2	0	0	2
295	2	1	2	1	2	0	1	0
296	2	1	2	1	2	0	1	1
297	2	1	2	1	2	0	1	2
298	2	1	2	1	2	1	0	0
299	2	1	2	1	2	1	0	1
300	2	1	2	1	2	1	0	2

#	n	ℓ	k	q	e	f	g	2S
301	2	1	2	1	2	1	1	0
302	2	1	2	1	2	1	1	1
303	2	1	2	1	2	1	1	2
304	2	1	2	1	3	0	0	0

#	n	ℓ	k	q	e	f	g	2S
305	2	1	2	1	3	0	0	1
306	2	1	2	1	3	0	0	2
307	2	1	2	1	3	0	1	0
308	2	1	2	1	3	0	1	1
309	2	1	2	1	3	0	1	2
310	2	1	2	1	3	1	0	0
311	2	1	2	1	3	1	0	1
312	2	1	2	1	3	1	0	2
313	2	1	2	1	3	1	1	0
314	2	1	2	1	3	1	1	1
315	2	1	2	1	3	1	1	2
316	2	1	2	2	1	0	0	0
317	2	1	2	2	1	0	0	1
318	2	1	2	2	1	0	0	2
319	2	1	2	2	1	0	1	0
320	2	1	2	2	1	0	1	1
321	2	1	2	2	1	0	1	2
322	2	1	2	2	1	0	2	0
323	2	1	2	2	1	0	2	1
324	2	1	2	2	1	0	2	2
325	2	1	2	2	2	0	0	0
326	2	1	2	2	2	0	0	1
327	2	1	2	2	2	0	0	2
328	2	1	2	2	2	0	1	0
329	2	1	2	2	2	0	1	1
330	2	1	2	2	2	0	1	2

#	n	ℓ	k	q	e	f	g	2S
331	2	1	2	2	2	0	2	0
332	2	1	2	2	2	0	2	1
333	2	1	2	2	2	0	2	2
334	2	1	2	2	2	1	0	0
335	2	1	2	2	2	1	0	1
336	2	1	2	2	2	1	0	2
337	2	1	2	2	2	1	1	0
338	2	1	2	2	2	1	1	1
339	2	1	2	2	2	1	1	2
340	2	1	2	2	2	1	2	0
341	2	1	2	2	2	1	2	1
342	2	1	2	2	2	1	2	2
343	2	1	2	2	3	0	0	0
344	2	1	2	2	3	0	0	1
345	2	1	2	2	3	0	0	2
346	2	1	2	2	3	0	1	0
347	2	1	2	2	3	0	1	1
348	2	1	2	2	3	0	1	2
349	2	1	2	2	3	0	2	0
350	2	1	2	2	3	0	2	1

#	n	ℓ	k	q	e	f	g	2S
351	2	1	2	2	3	0	2	2
352	2	1	2	2	3	1	0	0
353	2	1	2	2	3	1	0	1
354	2	1	2	2	3	1	0	2
355	2	1	2	2	3	1	1	0
356	2	1	2	2	3	1	1	1
357	2	1	2	2	3	1	1	2
358	2	1	2	2	3	1	2	0
359	2	1	2	2	3	1	2	1
360	2	1	2	2	3	1	2	2

#	n	ℓ	k	q	e	f	g	2S
361	2	1	3	0	1	0	0	0
362	2	1	3	0	1	0	0	1
363	2	1	3	0	1	0	0	2
364	2	1	3	0	1	0	0	3
365	2	1	3	0	2	0	0	0
366	2	1	3	0	2	0	0	1
367	2	1	3	0	2	0	0	2
368	2	1	3	0	2	0	0	3
369	2	1	3	0	2	1	0	0
370	2	1	3	0	2	1	0	1
371	2	1	3	0	2	1	0	2
372	2	1	3	0	2	1	0	3
373	2	1	3	0	3	0	0	0
374	2	1	3	0	3	0	0	1
375	2	1	3	0	3	0	0	2
376	2	1	3	0	3	0	0	3
377	2	1	3	0	3	1	0	0
378	2	1	3	0	3	1	0	1
379	2	1	3	0	3	1	0	2
380	2	1	3	0	3	1	0	3
381	2	1	3	1	1	0	0	0
382	2	1	3	1	1	0	0	1
383	2	1	3	1	1	0	0	2
384	2	1	3	1	1	0	0	3
385	2	1	3	1	1	0	1	0
386	2	1	3	1	1	0	1	1
387	2	1	3	1	1	0	1	2
388	2	1	3	1	1	0	1	3
389	2	1	3	1	2	0	0	0
390	2	1	3	1	2	0	0	1

#	n	ℓ	k	q	e	f	g	2S
391	2	1	3	1	2	0	0	2
392	2	1	3	1	2	0	0	3
393	2	1	3	1	2	0	1	0
394	2	1	3	1	2	0	1	1
395	2	1	3	1	2	0	1	2

#	n	ℓ	k	q	e	f	g	2S
396	2	1	3	1	2	0	1	3
397	2	1	3	1	2	1	0	0
398	2	1	3	1	2	1	0	1
399	2	1	3	1	2	1	0	2
400	2	1	3	1	2	1	0	3
401	2	1	3	1	2	1	1	0
402	2	1	3	1	2	1	1	1
403	2	1	3	1	2	1	1	2
404	2	1	3	1	2	1	1	3
405	2	1	3	1	3	0	0	0
406	2	1	3	1	3	0	0	1
407	2	1	3	1	3	0	0	2
408	2	1	3	1	3	0	0	3
409	2	1	3	1	3	0	1	0
410	2	1	3	1	3	0	1	1
411	2	1	3	1	3	0	1	2
412	2	1	3	1	3	0	1	3
413	2	1	3	1	3	1	0	0
414	2	1	3	1	3	1	0	1
415	2	1	3	1	3	1	0	2
416	2	1	3	1	3	1	0	3
417	2	1	3	1	3	1	1	0
418	2	1	3	1	3	1	1	1
419	2	1	3	1	3	1	1	2
420	2	1	3	1	3	1	1	3

#	n	ℓ	k	q	e	f	g	2S
421	2	1	3	2	1	0	0	0
422	2	1	3	2	1	0	0	1
423	2	1	3	2	1	0	0	2
424	2	1	3	2	1	0	0	3
425	2	1	3	2	1	0	1	0
426	2	1	3	2	1	0	1	1
427	2	1	3	2	1	0	1	2
428	2	1	3	2	1	0	1	3
429	2	1	3	2	1	0	2	0
430	2	1	3	2	1	0	2	1
431	2	1	3	2	1	0	2	2
432	2	1	3	2	1	0	2	3
433	2	1	3	2	2	0	0	0
434	2	1	3	2	2	0	0	1
435	2	1	3	2	2	0	0	2
436	2	1	3	2	2	0	0	3
437	2	1	3	2	2	0	1	0
438	2	1	3	2	2	0	1	1
439	2	1	3	2	2	0	1	2
440	2	1	3	2	2	0	1	3
441	2	1	3	2	2	0	2	0

#	n	ℓ	k	q	e	f	g	2S
442	2	1	3	2	2	0	2	1
443	2	1	3	2	2	0	2	2
444	2	1	3	2	2	0	2	3
445	2	1	3	2	2	1	0	0
446	2	1	3	2	2	1	0	1
447	2	1	3	2	2	1	0	2
448	2	1	3	2	2	1	0	3
449	2	1	3	2	2	1	1	0
450	2	1	3	2	2	1	1	1

#	n	ℓ	k	q	e	f	g	2S
451	2	1	3	2	2	1	1	2
452	2	1	3	2	2	1	1	3
453	2	1	3	2	2	1	2	0
454	2	1	3	2	2	1	2	1
455	2	1	3	2	2	1	2	2
456	2	1	3	2	2	1	2	3
457	2	1	3	2	3	0	0	0
458	2	1	3	2	3	0	0	1
459	2	1	3	2	3	0	0	2
460	2	1	3	2	3	0	0	3
461	2	1	3	2	3	0	1	0
462	2	1	3	2	3	0	1	1
463	2	1	3	2	3	0	1	2
464	2	1	3	2	3	0	1	3
465	2	1	3	2	3	0	2	0
466	2	1	3	2	3	0	2	1
467	2	1	3	2	3	0	2	2
468	2	1	3	2	3	0	2	3
469	2	1	3	2	3	1	0	0
470	2	1	3	2	3	1	0	1
471	2	1	3	2	3	1	0	2
472	2	1	3	2	3	1	0	3
473	2	1	3	2	3	1	1	0
474	2	1	3	2	3	1	1	1
475	2	1	3	2	3	1	1	2
476	2	1	3	2	3	1	1	3
477	2	1	3	2	3	1	2	0
478	2	1	3	2	3	1	2	1
479	2	1	3	2	3	1	2	2
480	2	1	3	2	3	1	2	3

#	n	ℓ	k	q	e	f	g	2S
481	2	1	3	3	1	0	0	0
482	2	1	3	3	1	0	0	1
483	2	1	3	3	1	0	0	2
484	2	1	3	3	1	0	0	3
485	2	1	3	3	1	0	1	0
486	2	1	3	3	1	0	1	1

#	n	ℓ	k	q	e	f	g	2S
487	2	1	3	3	1	0	1	2
488	2	1	3	3	1	0	1	3
489	2	1	3	3	1	0	2	0
490	2	1	3	3	1	0	2	1
491	2	1	3	3	1	0	2	2
492	2	1	3	3	1	0	2	3
493	2	1	3	3	2	0	0	0
494	2	1	3	3	2	0	0	1
495	2	1	3	3	2	0	0	2
496	2	1	3	3	2	0	0	3
497	2	1	3	3	2	0	1	0
498	2	1	3	3	2	0	1	1
499	2	1	3	3	2	0	1	2
500	2	1	3	3	2	0	1	3
501	2	1	3	3	2	0	2	0
502	2	1	3	3	2	0	2	1
503	2	1	3	3	2	0	2	2
504	2	1	3	3	2	0	2	3
505	2	1	3	3	2	1	0	0
506	2	1	3	3	2	1	0	1
507	2	1	3	3	2	1	0	2
508	2	1	3	3	2	1	0	3
509	2	1	3	3	2	1	1	0
510	2	1	3	3	2	1	1	1

#	n	ℓ	k	q	e	f	g	2S
511	2	1	3	3	2	1	1	2
512	2	1	3	3	2	1	1	3
513	2	1	3	3	2	1	2	0
514	2	1	3	3	2	1	2	1
515	2	1	3	3	2	1	2	2
516	2	1	3	3	2	1	2	3
517	2	1	3	3	2	1	3	0
518	2	1	3	3	2	1	3	1
519	2	1	3	3	2	1	3	2
520	2	1	3	3	2	1	3	3
521	2	1	3	3	3	0	0	0
522	2	1	3	3	3	0	0	1
523	2	1	3	3	3	0	0	2
524	2	1	3	3	3	0	0	3
525	2	1	3	3	3	0	1	0
526	2	1	3	3	3	0	1	1
527	2	1	3	3	3	0	1	2
528	2	1	3	3	3	0	1	3
529	2	1	3	3	3	0	2	0
530	2	1	3	3	3	0	2	1
531	2	1	3	3	3	0	2	2
532	2	1	3	3	3	0	2	3

#	n	ℓ	k	q	e	f	g	2S
533	2	1	3	3	3	1	0	0
534	2	1	3	3	3	1	0	1
535	2	1	3	3	3	1	0	2
536	2	1	3	3	3	1	0	3
537	2	1	3	3	3	1	1	0
538	2	1	3	3	3	1	1	1
539	2	1	3	3	3	1	1	2
540	2	1	3	3	3	1	1	3

#	n	ℓ	k	q	e	f	g	2S
541	2	1	3	3	3	1	2	0
542	2	1	3	3	3	1	2	1
543	2	1	3	3	3	1	2	2
544	2	1	3	3	3	1	2	3
545	2	1	3	3	3	1	3	0
546	2	1	3	3	3	1	3	1
547	2	1	3	3	3	1	3	2
548	2	1	3	3	3	1	3	3
549	3	0	1	0	1	0	0	0
550	3	0	1	0	1	0	0	1
551	3	0	1	0	2	0	0	0
552	3	0	1	0	2	0	0	1
553	3	0	1	0	2	1	0	0
554	3	0	1	0	2	1	0	1
555	3	0	1	0	3	0	0	0
556	3	0	1	0	3	0	0	1
557	3	0	1	0	3	1	0	0
558	3	0	1	0	3	1	0	1
559	3	0	1	1	1	0	0	0
560	3	0	1	1	1	0	0	1
561	3	0	1	1	1	0	1	0
562	3	0	1	1	1	0	1	1
563	3	0	1	1	2	0	0	0
564	3	0	1	1	2	0	0	1
565	3	0	1	1	2	0	1	0
566	3	0	1	1	2	0	1	1
567	3	0	1	1	2	1	0	0
568	3	0	1	1	2	1	0	1
569	3	0	1	1	2	1	1	0
570	3	0	1	1	2	1	1	1

#	n	ℓ	k	q	e	f	g	2S
571	3	0	1	1	3	0	0	0
572	3	0	1	1	3	0	0	1
573	3	0	1	1	3	0	1	0
574	3	0	1	1	3	0	1	1
575	3	0	1	1	3	1	0	0
576	3	0	1	1	3	1	0	1
577	3	0	1	1	3	1	1	0

#	n	ℓ	k	q	e	f	g	2S
578	3	0	1	1	3	1	1	1
579	3	0	2	0	1	0	0	0
580	3	0	2	0	1	0	0	1
581	3	0	2	0	1	0	0	2
582	3	0	2	0	2	0	0	0
583	3	0	2	0	2	0	0	1
584	3	0	2	0	2	0	0	2
585	3	0	2	0	2	1	0	0
586	3	0	2	0	2	1	0	1
587	3	0	2	0	2	1	0	2
588	3	0	2	0	3	0	0	0
589	3	0	2	0	3	0	0	1
590	3	0	2	0	3	0	0	2
591	3	0	2	0	3	1	0	0
592	3	0	2	0	3	1	0	1
593	3	0	2	0	3	1	0	2
594	3	0	2	1	1	0	0	0
595	3	0	2	1	1	0	0	1
596	3	0	2	1	1	0	0	2
597	3	0	2	1	1	0	1	0
598	3	0	2	1	1	0	1	1
599	3	0	2	1	1	0	1	2
600	3	0	2	1	2	0	0	0

#	n	ℓ	k	q	e	f	g	2S
601	3	0	2	1	2	0	0	1
602	3	0	2	1	2	0	0	2
603	3	0	2	1	2	0	1	0
604	3	0	2	1	2	0	1	1
605	3	0	2	1	2	0	1	2
606	3	0	2	1	2	1	0	0
607	3	0	2	1	2	1	0	1
608	3	0	2	1	2	1	0	2
609	3	0	2	1	2	1	1	0
610	3	0	2	1	2	1	1	1
611	3	0	2	1	2	1	1	2
612	3	0	2	1	3	0	0	0
613	3	0	2	1	3	0	0	1
614	3	0	2	1	3	0	0	2
615	3	0	2	1	3	0	1	0
616	3	0	2	1	3	0	1	1
617	3	0	2	1	3	0	1	2
618	3	0	2	1	3	1	0	0
619	3	0	2	1	3	1	0	1
620	3	0	2	1	3	1	0	2
621	3	0	2	1	3	1	1	0
622	3	0	2	1	3	1	1	1
623	3	0	2	1	3	1	1	2

#	n	ℓ	k	q	e	f	g	2S
624	3	0	2	2	1	0	0	0
625	3	0	2	2	1	0	0	1
626	3	0	2	2	1	0	0	2
627	3	0	2	2	1	0	1	0
628	3	0	2	2	1	0	1	1
629	3	0	2	2	1	0	1	2
630	3	0	2	2	1	0	2	0

#	n	ℓ	k	q	e	f	g	2S
631	3	0	2	2	1	0	2	1
632	3	0	2	2	1	0	2	2
633	3	0	2	2	2	0	0	0
634	3	0	2	2	2	0	0	1
635	3	0	2	2	2	0	0	2
636	3	0	2	2	2	0	1	0
637	3	0	2	2	2	0	1	1
638	3	0	2	2	2	0	1	2
639	3	0	2	2	2	0	2	0
640	3	0	2	2	2	0	2	1
641	3	0	2	2	2	0	2	2
642	3	0	2	2	2	1	0	0
643	3	0	2	2	2	1	0	1
644	3	0	2	2	2	1	0	2
645	3	0	2	2	2	1	1	0
646	3	0	2	2	2	1	1	1
647	3	0	2	2	2	1	1	2
648	3	0	2	2	2	1	2	0
649	3	0	2	2	2	1	2	1
650	3	0	2	2	2	1	2	2
651	3	0	2	2	3	0	0	0
652	3	0	2	2	3	0	0	1
653	3	0	2	2	3	0	0	2
654	3	0	2	2	3	0	1	0
655	3	0	2	2	3	0	1	1
656	3	0	2	2	3	0	1	2
657	3	0	2	2	3	0	2	0
658	3	0	2	2	3	0	2	1
659	3	0	2	2	3	0	2	2
660	3	0	2	2	3	1	0	0

#	n	ℓ	k	q	e	f	g	2S
661	3	0	2	2	3	1	0	1
662	3	0	2	2	3	1	0	2
663	3	0	2	2	3	1	1	0
664	3	0	2	2	3	1	1	1
665	3	0	2	2	3	1	1	2
666	3	0	2	2	3	1	2	0
667	3	0	2	2	3	1	2	1
668	3	0	2	2	3	1	2	2

#	n	ℓ	k	q	e	f	g	2S
669	3	1	1	0	1	0	0	0
670	3	1	1	0	1	0	0	1
671	3	1	1	0	2	0	0	0
672	3	1	1	0	2	0	0	1
673	3	1	1	0	2	1	0	0
674	3	1	1	0	2	1	0	1
675	3	1	1	0	3	0	0	0
676	3	1	1	0	3	0	0	1
677	3	1	1	0	3	1	0	0
678	3	1	1	0	3	1	0	1
679	3	1	1	1	1	0	0	0
680	3	1	1	1	1	0	0	1
681	3	1	1	1	1	0	1	0
682	3	1	1	1	1	0	1	1
683	3	1	1	1	2	0	0	0
684	3	1	1	1	2	0	0	1
685	3	1	1	1	2	0	1	0
686	3	1	1	1	2	0	1	1
687	3	1	1	1	2	1	0	0
688	3	1	1	1	2	1	0	1
689	3	1	1	1	2	1	1	0
690	3	1	1	1	2	1	1	1

#	n	ℓ	k	q	e	f	g	2S
691	3	1	1	1	3	0	0	0
692	3	1	1	1	3	0	0	1
693	3	1	1	1	3	0	1	0
694	3	1	1	1	3	0	1	1
695	3	1	1	1	3	1	0	0
696	3	1	1	1	3	1	0	1
697	3	1	1	1	3	1	1	0
698	3	1	1	1	3	1	1	1
699	3	1	2	0	1	0	0	0
700	3	1	2	0	1	0	0	1
701	3	1	2	0	1	0	0	2
702	3	1	2	0	2	0	0	0
703	3	1	2	0	2	0	0	1
704	3	1	2	0	2	0	0	2
705	3	1	2	0	2	1	0	0
706	3	1	2	0	2	1	0	1
707	3	1	2	0	2	1	0	2
708	3	1	2	0	3	0	0	0
709	3	1	2	0	3	0	0	1
710	3	1	2	0	3	0	0	2
711	3	1	2	0	3	1	0	0
712	3	1	2	0	3	1	0	1
713	3	1	2	0	3	1	0	2
714	3	1	2	1	1	0	0	0

#	n	ℓ	k	q	e	f	g	2S
715	3	1	2	1	1	0	0	1
716	3	1	2	1	1	0	0	2
717	3	1	2	1	1	0	1	0
718	3	1	2	1	1	0	1	1
719	3	1	2	1	1	0	1	2
720	3	1	2	1	2	0	0	0

#	n	ℓ	k	q	e	f	g	2S
721	3	1	2	1	2	0	0	1
722	3	1	2	1	2	0	0	2
723	3	1	2	1	2	0	1	0
724	3	1	2	1	2	0	1	1
725	3	1	2	1	2	0	1	2
726	3	1	2	1	2	1	0	0
727	3	1	2	1	2	1	0	1
728	3	1	2	1	2	1	0	2
729	3	1	2	1	2	1	1	0
730	3	1	2	1	2	1	1	1
731	3	1	2	1	2	1	1	2
732	3	1	2	1	3	0	0	0
733	3	1	2	1	3	0	0	1
734	3	1	2	1	3	0	0	2
735	3	1	2	1	3	0	1	0
736	3	1	2	1	3	0	1	1
737	3	1	2	1	3	0	1	2
738	3	1	2	1	3	1	0	0
739	3	1	2	1	3	1	0	1
740	3	1	2	1	3	1	0	2
741	3	1	2	1	3	1	1	0
742	3	1	2	1	3	1	1	1
743	3	1	2	1	3	1	1	2
744	3	1	2	2	1	0	0	0
745	3	1	2	2	1	0	0	1
746	3	1	2	2	1	0	0	2
747	3	1	2	2	1	0	1	0
748	3	1	2	2	1	0	1	1
749	3	1	2	2	1	0	1	2
750	3	1	2	2	1	0	2	0

#	n	ℓ	k	q	e	f	g	2S
751	3	1	2	2	1	0	2	1
752	3	1	2	2	1	0	2	2
753	3	1	2	2	2	0	0	0
754	3	1	2	2	2	0	0	1
755	3	1	2	2	2	0	0	2
756	3	1	2	2	2	0	1	0
757	3	1	2	2	2	0	1	1
758	3	1	2	2	2	0	1	2
759	3	1	2	2	2	0	2	0

#	n	ℓ	k	q	e	f	g	2S
760	3	1	2	2	2	0	2	1
761	3	1	2	2	2	0	2	2
762	3	1	2	2	2	1	0	0
763	3	1	2	2	2	1	0	1
764	3	1	2	2	2	1	0	2
765	3	1	2	2	2	1	1	0
766	3	1	2	2	2	1	1	1
767	3	1	2	2	2	1	1	2
768	3	1	2	2	2	1	2	0
769	3	1	2	2	2	1	2	1
770	3	1	2	2	2	1	2	2
771	3	1	2	2	3	0	0	0
772	3	1	2	2	3	0	0	1
773	3	1	2	2	3	0	0	2
774	3	1	2	2	3	0	1	0
775	3	1	2	2	3	0	1	1
776	3	1	2	2	3	0	1	2
777	3	1	2	2	3	0	2	0
778	3	1	2	2	3	0	2	1
779	3	1	2	2	3	0	2	2
780	3	1	2	2	3	1	0	0

#	n	ℓ	k	q	e	f	g	2S
781	3	1	2	2	3	1	0	1
782	3	1	2	2	3	1	0	2
783	3	1	2	2	3	1	1	0
784	3	1	2	2	3	1	1	1
785	3	1	2	2	3	1	1	2
786	3	1	2	2	3	1	2	0
787	3	1	2	2	3	1	2	1
788	3	1	2	2	3	1	2	2
789	3	1	3	0	1	0	0	0
790	3	1	3	0	1	0	0	1
791	3	1	3	0	1	0	0	2
792	3	1	3	0	1	0	0	3
793	3	1	3	0	2	0	0	0
794	3	1	3	0	2	0	0	1
795	3	1	3	0	2	0	0	2
796	3	1	3	0	2	0	0	3
797	3	1	3	0	2	1	0	0
798	3	1	3	0	2	1	0	1
799	3	1	3	0	2	1	0	2
800	3	1	3	0	2	1	0	3
801	3	1	3	0	3	0	0	0
802	3	1	3	0	3	0	0	1
803	3	1	3	0	3	0	0	2
804	3	1	3	0	3	0	0	3
805	3	1	3	0	3	1	0	0

#	n	ℓ	k	q	e	f	g	2S
806	3	1	3	0	3	1	0	1
807	3	1	3	0	3	1	0	2
808	3	1	3	0	3	1	0	3
809	3	1	3	1	1	0	0	0
810	3	1	3	1	1	0	0	1

#	n	ℓ	k	q	e	f	g	2S
811	3	1	3	1	1	0	0	2
812	3	1	3	1	1	0	0	3
813	3	1	3	1	1	0	1	0
814	3	1	3	1	1	0	1	1
815	3	1	3	1	1	0	1	2
816	3	1	3	1	1	0	1	3
817	3	1	3	1	2	0	0	0
818	3	1	3	1	2	0	0	1
819	3	1	3	1	2	0	0	2
820	3	1	3	1	2	0	0	3
821	3	1	3	1	2	0	1	0
822	3	1	3	1	2	0	1	1
823	3	1	3	1	2	0	1	2
824	3	1	3	1	2	0	1	3
825	3	1	3	1	2	1	0	0
826	3	1	3	1	2	1	0	1
827	3	1	3	1	2	1	0	2
828	3	1	3	1	2	1	0	3
829	3	1	3	1	2	1	1	0
830	3	1	3	1	2	1	1	1
831	3	1	3	1	2	1	1	2
832	3	1	3	1	2	1	1	3
833	3	1	3	1	3	0	0	0
834	3	1	3	1	3	0	0	1
835	3	1	3	1	3	0	0	2
836	3	1	3	1	3	0	0	3
837	3	1	3	1	3	0	1	0
838	3	1	3	1	3	0	1	1
839	3	1	3	1	3	0	1	2
840	3	1	3	1	3	0	1	3

#	n	ℓ	k	q	e	f	g	2S
841	3	1	3	1	3	1	0	0
842	3	1	3	1	3	1	0	1
843	3	1	3	1	3	1	0	2
844	3	1	3	1	3	1	0	3
845	3	1	3	1	3	1	1	0
846	3	1	3	1	3	1	1	1
847	3	1	3	1	3	1	1	2
848	3	1	3	1	3	1	1	3
849	3	1	3	2	1	0	0	0
850	3	1	3	2	1	0	0	1

#	n	ℓ	k	q	e	f	g	2S
851	3	1	3	2	1	0	0	2
852	3	1	3	2	1	0	0	3
853	3	1	3	2	1	0	1	0
854	3	1	3	2	1	0	1	1
855	3	1	3	2	1	0	1	2
856	3	1	3	2	1	0	1	3
857	3	1	3	2	1	0	2	0
858	3	1	3	2	1	0	2	1
859	3	1	3	2	1	0	2	2
860	3	1	3	2	1	0	2	3
861	3	1	3	2	2	0	0	0
862	3	1	3	2	2	0	0	1
863	3	1	3	2	2	0	0	2
864	3	1	3	2	2	0	0	3
865	3	1	3	2	2	0	1	0
866	3	1	3	2	2	0	1	1
867	3	1	3	2	2	0	1	2
868	3	1	3	2	2	0	1	3
869	3	1	3	2	2	0	2	0
870	3	1	3	2	2	0	2	1

#	n	ℓ	k	q	e	f	g	2S
871	3	1	3	2	2	0	2	2
872	3	1	3	2	2	0	2	3
873	3	1	3	2	2	1	0	0
874	3	1	3	2	2	1	0	1
875	3	1	3	2	2	1	0	2
876	3	1	3	2	2	1	0	3
877	3	1	3	2	2	1	1	0
878	3	1	3	2	2	1	1	1
879	3	1	3	2	2	1	1	2
880	3	1	3	2	2	1	1	3
881	3	1	3	2	2	1	2	0
882	3	1	3	2	2	1	2	1
883	3	1	3	2	2	1	2	2
884	3	1	3	2	2	1	2	3
885	3	1	3	2	3	0	0	0
886	3	1	3	2	3	0	0	1
887	3	1	3	2	3	0	0	2
888	3	1	3	2	3	0	0	3
889	3	1	3	2	3	0	1	0
890	3	1	3	2	3	0	1	1
891	3	1	3	2	3	0	1	2
892	3	1	3	2	3	0	1	3
893	3	1	3	2	3	0	2	0
894	3	1	3	2	3	0	2	1
895	3	1	3	2	3	0	2	2
896	3	1	3	2	3	0	2	3

#	n	ℓ	k	q	e	f	g	2S
897	3	1	3	2	3	1	0	0
898	3	1	3	2	3	1	0	1
899	3	1	3	2	3	1	0	2
900	3	1	3	2	3	1	0	3

#	n	ℓ	k	q	e	f	g	2S
901	3	1	3	2	3	1	1	0
902	3	1	3	2	3	1	1	1
903	3	1	3	2	3	1	1	2
904	3	1	3	2	3	1	1	3
905	3	1	3	2	3	1	2	0
906	3	1	3	2	3	1	2	1
907	3	1	3	2	3	1	2	2
908	3	1	3	2	3	1	2	3
909	3	1	3	3	1	0	0	0
910	3	1	3	3	1	0	0	1
911	3	1	3	3	1	0	0	2
912	3	1	3	3	1	0	0	3
913	3	1	3	3	1	0	1	0
914	3	1	3	3	1	0	1	1
915	3	1	3	3	1	0	1	2
916	3	1	3	3	1	0	1	3
917	3	1	3	3	1	0	2	0
918	3	1	3	3	1	0	2	1
919	3	1	3	3	1	0	2	2
920	3	1	3	3	1	0	2	3
921	3	1	3	3	2	0	0	0
922	3	1	3	3	2	0	0	1
923	3	1	3	3	2	0	0	2
924	3	1	3	3	2	0	0	3
925	3	1	3	3	2	0	1	0
926	3	1	3	3	2	0	1	1
927	3	1	3	3	2	0	1	2
928	3	1	3	3	2	0	1	3
929	3	1	3	3	2	0	2	0
930	3	1	3	3	2	0	2	1

#	n	ℓ	k	q	e	f	g	2S
931	3	1	3	3	2	0	2	2
932	3	1	3	3	2	0	2	3
933	3	1	3	3	2	1	0	0
934	3	1	3	3	2	1	0	1
935	3	1	3	3	2	1	0	2
936	3	1	3	3	2	1	0	3
937	3	1	3	3	2	1	1	0
938	3	1	3	3	2	1	1	1
939	3	1	3	3	2	1	1	2
940	3	1	3	3	2	1	1	3
941	3	1	3	3	2	1	2	0

#	n	ℓ	k	q	e	f	g	2S
942	3	1	3	3	2	1	2	1
943	3	1	3	3	2	1	2	2
944	3	1	3	3	2	1	2	3
945	3	1	3	3	2	1	3	0
946	3	1	3	3	2	1	3	1
947	3	1	3	3	2	1	3	2
948	3	1	3	3	2	1	3	3
949	3	1	3	3	3	0	0	0
950	3	1	3	3	3	0	0	1
951	3	1	3	3	3	0	0	2
952	3	1	3	3	3	0	0	3
953	3	1	3	3	3	0	1	0
954	3	1	3	3	3	0	1	1
955	3	1	3	3	3	0	1	2
956	3	1	3	3	3	0	1	3
957	3	1	3	3	3	0	2	0
958	3	1	3	3	3	0	2	1
959	3	1	3	3	3	0	2	2
960	3	1	3	3	3	0	2	3

#	n	ℓ	k	q	e	f	g	2S
961	3	1	3	3	3	1	0	0
962	3	1	3	3	3	1	0	1
963	3	1	3	3	3	1	0	2
964	3	1	3	3	3	1	0	3
965	3	1	3	3	3	1	1	0
966	3	1	3	3	3	1	1	1
967	3	1	3	3	3	1	1	2
968	3	1	3	3	3	1	1	3
969	3	1	3	3	3	1	2	0
970	3	1	3	3	3	1	2	1
971	3	1	3	3	3	1	2	2
972	3	1	3	3	3	1	2	3
973	3	1	3	3	3	1	3	0
974	3	1	3	3	3	1	3	1
975	3	1	3	3	3	1	3	2
976	3	1	3	3	3	1	3	3

976 rows. $F(1) = 976$, $F(-1) = 2$, $E = 0$, density 0.1412 of a box of 6,912.

The periodic table — all 90 cells, (period, group)

period	groups occupied
1	1, 18
2	1, 2, 13, 14, 15, 16, 17, 18
3	1, 2, 13, 14, 15, 16, 17, 18
4	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18
5	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18
6	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18

period	groups occupied
7	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18

90 cells in a box of 126. $E = 36$ — and the 36 are the cells the coordinates admit and no element occupies, which must be supplied from outside.

Janet — all 118, $(n+\ell, Z)$

$n+\ell$	atomic numbers
1	1–2 (2)
2	3–4 (2)
3	5–12 (8)
4	13–20 (8)
5	21–38 (18)
6	39–56 (18)
7	57–88 (32)
8	89–118 (30)

118 cells, $E = 0$. The same 118 elements as the table above, reordered by $n+\ell$.

The calendar — 365 cells, (month, day)

month	days	month	days
January	31	July	31
February	28	August	31
March	31	September	30
April	30	October	31
May	31	November	30
June	30	December	31

365 cells in a box of 372. $E = 7$ — February's missing 30th and 31st, and the four thirty-day months' 31sts. **$365 = 5 \cdot 73$, and 73 exceeds every rung**, so it is genuinely constrained.

A box ordering — all 35, $l \geq w \geq h$

$l \geq w \geq h$
0-0-0 1-0-0 1-1-0 1-1-1 2-0-0 2-1-0 2-1-1
2-2-0 2-2-1 2-2-2 3-0-0 3-1-0 3-1-1 3-2-0
3-2-1 3-2-2 3-3-0 3-3-1 3-3-2 3-3-3 4-0-0
4-1-0 4-1-1 4-2-0 4-2-1 4-2-2 4-3-0 4-3-1
4-3-2 4-3-3 4-4-0 4-4-1 4-4-2 4-4-3 4-4-4

35 cells in a box of 125, $E = 0$. $35 = 5 \cdot 7$ and 7 exceeds the rung of 5: a real constraint.

VIII • WHAT EACH INDEX IS, AND WHAT ONLY IT CONTRIBUTES

Λ is the only index here built from physics. The rest are built from Λ , from the act of building Λ , or from drawing the same subject another way. *This section states for each one: what it is, where it comes from in the published record, and the one thing it contributes to the atomic index that no other index can.*

Λ — the atomic index

What it is. Eight integer coordinates ($n, \ell, k, q, e, f, g, 2S$) under eight inequalities, 976 cells, $E = 0$.

Where it comes from. The constraints are not this work's: $\ell \leq n-1$ is the hydrogen solution (Bohr 1913; Schrödinger 1926); $k \leq 4\ell+2$ is Stoner's subshell capacity (1924) made exclusive by Pauli (1925); $2S \leq k$ is Pauli with Hund's first rule (1925). Λ **assembles them; it introduces none.**

What only it contributes. *The transfer.* Λ is the only index here whose cells are MOVES rather than positions — a cell carries a source subshell, a target subshell and a count transferred. **Everything about composition, time columns and categories follows from that and from nothing else in this compendium.**

The tower $\Lambda_8 \dots \Lambda_{13}$

What it is. Λ with the coupling quantum numbers adjoined one at a time — seniority, the core J, K, then J. Cell counts 976, 1654, 2535, 13585, 70905, 199130, $E = 0$ at every stage.

Where it comes from. Each axis is Racah or Condon & Shortley: seniority is Racah, *Theory of complex spectra III*, Phys. Rev. **63** (1943) 367–382; the core J and the recoupling coefficients are Racah, Phys. Rev. **62** (1942) 438–462; the jK and LK schemes are Condon & Shortley (1935), ch. X.

What only it contributes. *The price of an axis.* The tower is the only construction here that adds one coordinate at a time and measures what each costs and buys — **cycle rank, composability, exactness, and the point at which the tree becomes a graph.** No single index can show that; it takes a sequence.

The periodic table — (period, group)

What it is. 90 cells in a box of 126, $E = 36$.

Where it comes from. Mendeleev, *Über die Beziehungen der Eigenschaften zu den Atomgewichten der Elemente**, Z. Chem. **12** (1869) 405–406. *At least a thousand periodic systems have been published since (van Spronsen; Scerri, A Tale of Seven Scientists**, 2016).

What only it contributes. *The control.* It is the one index whose coordinates were fixed by other people for other reasons, long before this work, and which nonetheless supplies **118 elements and 18,288 constraint tests with zero failures. An index built by the author cannot be a control; this one can.**

Janet's left-step table — ($n+\ell, Z$)

What it is. The same 118 elements, ordered on $n+\ell$. 118 cells in a box of 944, $E = 0$.

Where it comes from. Charles Janet, *Considérations sur la structure du noyau de l'atome**, Beauvais (1929), with the table first published in 1928. Janet recognised the ($n+\ell$) rule before Madelung, who arrived at it around 1926 and did not publish until 1936.* The shell-length sequence 2, 8, 8, 18, 18, 32, 32 follows from the rule by the Klechkovski–Hakala formulas.

What only it contributes. *That E is coordinate-relative, demonstrated on one subject rather than argued.* The periodic table and Janet index THE SAME 118 elements and give $E = 36$ and $E = 0$. **No pair of indexes anywhere else in this work makes that point as cleanly, because no other pair shares its subject exactly.**

And an open problem sits underneath it. *Why* the $n+\ell$ ordering holds is unresolved — Löwdin's challenge, still open; see Allen & Knight, The Löwdin challenge: origin of the $n+l, n$ (Madelung) rule*, Int. J. Quantum Chem. **90** (2003) 80–88. This work uses the ordering and does not explain it, and the orbital-collapse result (registers 1187–1190) is a measurement against Janet's boundaries, not a derivation of them.*

The nuclide chart — (Z, N)

What it is. The measured nuclides indexed by proton and neutron count, $E = 9$.

Where it comes from. The chart is Segrè's, in use since the 1940s; the values are the AME2020 evaluation.

What only it contributes. *A defect whose cells are identifiable physics.* The nine cells are the mass formula's pairing and clustering terms. **It is the only index here where $E > 0$ and every missing cell can be named,** which is what makes the defect a measurement rather than a score.

The calendar — (month, day)

What it is. 365 cells in a box of 372, $E = 7$.

Where it comes from. The Gregorian reform of 1582, on the Julian arrangement of 46 BC. *The month lengths are a political inheritance, not a natural one.*

What only it contributes. *A subject with no physics in it at all.* The calendar shows the operator working on an object whose constraints are entirely conventional, **and $365 = 5 \cdot 73$ with $E = 7$ is the price of keeping January first.** No other index here separates the method from the physics so completely.

The box ordering and the chessboard

What they are. $\ell \geq w \geq h$ over five values: 35 cells in 125, $E = 0$. And (rank, file): 64 cells in 64, $E = 0$.

What only they contribute. *The floor and the ceiling.* The chessboard is a full product — $E = 0$ because there is no constraint at all. The box ordering is a genuine constraint that still closes. **Between them they show that $E = 0$ carries information only when the ambient box exceeds the cells,** which is why the compendium reports box alongside every defect.

The electromagnetic quotient

What it is. Λ_9 under the dipole selection rules: $|\Delta\ell| = 1$ and $\Delta S = 0$.

Where it comes from. The parity rule is Laporte, Z. Phys. **23** (1924) 135; the spin rule is Russell & Saunders, Astrophys. J. **61** (1925) 38; both have their group-theoretic ground in Wigner, Z. Phys. **43** (1927) 624.

What only it contributes. *That a selection rule is a QUOTIENT and not an extension.* Adjoining the multipole and ΔS as coordinates gives $E = 3$; taking the image gives $E = 0$. **It is the only index here that demonstrates the difference between adding a coordinate and dividing by one.**

Λ_{spectra} — the channel index

What it is. (Z , charge, ℓ , $2S+1$) over 104,832 cells, with a defect for each.

Where it comes from. Quantum defect theory is Seaton, MNRAS **118** (1958) 504–518, and Rep. Prog. Phys. **46** (1983) 167–257. *A published survey of the same object exists*: Theodosiou, Inokuti & Manson, At. Data Nucl. Data Tables **35** (1986) 473–486, Hartree–Slater, for all ionisation stages of all ions with $Z \leq 50$.

What only it contributes. *Values.* Every other index here holds cells and asks whether they close. **Λ_{spectra} holds cells that carry NUMBERS,** which is why it is the only index on which the Method equation's metric half — E_W — has anything to measure at all (registers 1196–1199).

The violation index

What it is. The companion paper's object over nine and fifteen letters, with a core of three conditions whose joint failure is irreducible.

Where it comes from. The conditions are physics — the null energy condition (Penrose 1965), ghost states (Pais & Uhlenbeck 1950), the equations of motion. *The core is a MINIMAL UNSATISFIABLE SUBSET* in the sense of Chinneck & Dravnieks,* Locating minimal infeasible constraint sets in linear programs*, ORSA J. Comput. **3** (1991) 157–168.

What only it contributes. *An index this work reasons about without holding.* Its cells are not printed anywhere. **It is the only test of whether the method says anything when the object is out of reach,** and the answer is that it locates a three-condition core from published summary numbers alone.

Λ_{phys} — the parameter index

What it is. Every number the work takes from physics, on four coordinates — kind (exact $\rightarrow$ fitted), source (mathematics $\rightarrow$ this work), domain (universal $\rightarrow$ one species), and how many registered objects rest on it. **22 parameters.**

Where it comes from. The parameters are Rydberg 1890, Bohr 1913, Stoner 1924, Pauli 1925, Hund 1925, Fermi 1928, Hartree 1928, Janet 1928, Mayer & Mayer 1933, Goeppert-Mayer 1941, Seaton 1958, Edlén 1964, Griffin, Andrew & Cowan

1969, CODATA 2018 — and eight numbers fitted here. *The Physics Compendium holds them in full, each with a definition, a dated source and a failure mode.*

What only it contributes. *Where the work would break first.* Every other index here asks whether a set of cells closes.

Λ_{phys} asks what the whole construction rests on, and the answer is a shape: eight of the twenty-two parameters are this work's own, **none of them is universal**, seven of the eight hold only in a region or for one species — and the three carrying the most objects are all READ from published tables rather than fitted: the ionisation limit (25 objects), the aufbau ordering (19), the Janet block boundary (15).

And three of its failure modes are MEASURED rather than anticipated — Seaton's ratio at 1.25 where the dipole term gives 1.00, the Thomas–Fermi exponent rising from 0.84 to 1.52 with charge against a single fitted 0.494, and the exchange coefficient whose fitted sign is opposite to the measured one.

The languages

What they are. Six readings of one index — order, analysis, algebra, geometry, information, statistics — with documentary as a seventh that has no operator.

Where they come from. Each operator is standard in its own field: closure from Moore (1910), max-entropy on marginals from Deming & Stephan (1940) and Csiszár (1975), conditional independence from Dawid (1979).

What only they contribute. *That translation is re-coordination and E is its cost.* **And the agreement theorem: $E(X) = 0$ if and only if the languages agree** — six indexes, three operators, no exception (register 1176).

The book's own indexes

What they are. The book at chapter resolution ($E = 578$) and at part resolution ($E = 0$); the register ($E = 6$ at 68.8% density); the reference index (38 cells, $E = 0$, and not a tree).

What only they contribute. *The method self-applied.* **This is the only place where $E > 0$ is a theorem about SHAPE rather than a gap in collection** — the book's chapters do not close because chapters are not a coordinate system, and saying so with a number is the point.

The one-line summary

index	the one thing only it gives
Λ	the transfer — cells that are moves
the tower	the price of an axis, measured one at a time
the periodic table	a control the author did not build
Janet	E is coordinate-relative, on one subject
the nuclide chart	a defect whose every cell is nameable physics
the calendar	a subject with no physics in it
box and chessboard	the floor and ceiling of what $E = 0$ means
the EM quotient	quotient versus extension
Λ_{spectra}	values, so the metric defect has something to measure
the violation index	reasoning about an index without holding it
Λ_{phys}	where the work would break first
the languages	translation as re-coordination
the book's own	$E > 0$ as a theorem about shape

Λ is the only one built from physics. The rest are built from Λ , from the act of building Λ , or from drawing the same subject another way — and that is the claim: an index is a coordinate system, and the coordinates come from the subject or from nowhere.

IX • THE INDEXES BUILT IN THE LÖWDIN WORK

Registers 1249–1429. Fifteen indexes: fourteen closed — Λ_{ladder} now among them — and Λ_{spectra} closing at the limit with eleven named cells. Each is stated the same way: what it indexes, its coordinates, the axis order that closes it, what it refuses, and the one thing it contributes.

Λ_{law} — the laws of the channel equation

What it is. Seven laws on (law, carrier). $E = 0$ at carrier order $u < p < \ell - \ell_{\text{core}} < Z - T$, and 288 of 120,960 orderings reach it.

What only it contributes. *The carrier.* The order is monotone in how LOCAL the variable is — u is the whole atom, p counts one ℓ 's shells, $\ell - \ell_{\text{core}}$ compares two angular momenta, $Z - T$ is one distance to one threshold. **The index says which law runs in which variable, and fitting a law in the wrong carrier is what took the ℓ -spread from a spurious r^2 0.868 to a real 0.356.**

Λ_{const} — the constants

What it is. Fourteen constants on (role, carrier), role order **exponent < centre < width < scale**. $E = 0$ at 2 of 576 orderings.

What only it contributes. *That standing cannot be a coordinate.* With attributed/derived/measured/fitted on an axis the index cannot close at any ordering; without it, it closes at once. **An index whose coordinates mix the object with the observer cannot close, because the observer's axis has no order the object respects.** The same fault recurs as `origin` in $\Lambda_{\text{var, kind}}$ in Λ_{phys} and `state` in Λ_{ladder} .

And the role axis orders by how much physics a number has absorbed. Every CENTRE in the system is a small integer or half-integer — 2, 2, 2.5, −1.5, 5.4 — and no SCALE is.

Λ_{var} — the variables

What it is. Twelve variables on (body, role), body order **nucleus < core < core+rydberg < nucleus+core < rydberg**. $E = 0$.

What only it contributes. *That there is no nucleus + rydberg cell.* Not one variable relates them directly: every quantity connecting the Rydberg electron to the nucleus — c , u — is a nucleus–core quantity that the Rydberg electron then reads. **That is the screening statement appearing as a structural fact of the index rather than a modelling choice, and it is the three-body factorisation: nucleus–core gives c and u , core–Rydberg gives p , n_0 and T , and the Rydberg electron's own ℓ closes it.**

And the singleton rule. Λ_{var} passes $\mathcal{R}$ with both δ and n^* present, because both land in the same cell — **only a second criterion sees it. An index is closed when its output class is a singleton; two outputs mean the observer is still choosing which to read.**

Λ_{ryd} — the Rydberg series

What it is. Five cells on (Ritz order, ℓ , sign). $E = 0$ once δ_2 is reclassified as INTERMEDIATE — it is fitted from the series, never measured.

What only it contributes. *The n -dependence Λ_{spectra} discards.* A channel's δ is not one number but a convergent sequence — Cd I's ns series gives 3.7171, 3.6835, 3.6719, 3.6665, 3.6637, 3.6621, 3.6610. Λ_{spectra} holds δ_0 ; Λ_{ryd} holds δ_2 and δ_4 .

And the sign of δ_2 is penetration, not ℓ : 0 of 3 positive at $p = 0$, 2 of 2 at $p = 5$. Seaton's ratio is valid where $p = 0$ at 1.15 ± 0.21 and UNDEFINED where $p \geq 1$ — a domain, not a failure.

Λ_{charge} — the roles of the charge

What it is. Nine occurrences on (role, carrier, sign, regime), nine distinct cells, $E = 0$, and dropping any single role also gives 0.

What only it contributes. *That c is three coordinates in one symbol.* The charge appears inside $u = \ln(N_e/c^{(2/3)})$, as the base of $c^{(-x)}$, and inside the exponent x itself. **Every degeneracy of the session was between a charge term and something**

else, and this index says why: one symbol occupying three positions, and any fit letting two of them float will trade them.

A warning the index also carries: declaring the three roles as coordinates in Λ_{spectra} raises E from 1929 to 13,605. A closed sub-index says its roles are well-posed; it says nothing about whether the parent index wants them.

Λ_{cross} — the crossing values

What it is. The value at which an ordering flips, $a_{\text{cross}} = \Delta n(\sqrt{p_g} + \sqrt{p_r})/(p_g - p_r)$ — nineteen distinct surds across the table, including $1/\sqrt{3}$, $1/\sqrt{2}$, 1, $1/(\sqrt{3}-1)$, $1+1/\sqrt{2}$ and $1/(\sqrt{2}-1)$. $E = 0$ on $(\Delta n, \ell_g)$.

What only it contributes. *An index with no statistics language.* Order, algebra (closed in $\mathbb{Q}(\sqrt{p})$ for $p \leq 7$) and geometry all speak of it; **analysis and statistics do not.** A quantity that is ordinal, algebraic and geometric but not analytic is an INTEGER OBJECT: it cannot be fitted and needs no fitting.

And it is the one index whose unfixable cell is fixed anyway — a node count needs no measurement. **Nine of eleven unfixable cells across the work are expectation values $\langle \Psi | \hat{O} | \Psi \rangle$; Λ_{cross} is the exception, and it is the only index with no statistics language. The two facts are the same fact.**

Λ_{descent} — the charge dependence

What it is. $a(c)$ on (c, regime) . $E = 0$.

What only it contributes. *The only place a fit belongs.* Order and analysis both speak of it, information says c adds everything, and statistics answers yes. **Every fit of the session was applied to the WHOLE of a — the crossing and the descent together — and that is why the constants kept absorbing each other: half the object has no analysis language, and the fitted parameters were competing to represent a set of surds.**

Λ_{phys} — the parameters, rebuilt

What it is. Twenty-one parameters of real atoms on $(\text{source}, \text{domain})$, $E = 0$, at source order **standard** < **mathematics** < **literature** < **this work** and domain order **universal** < **all elements** < **a region** < **one species**.

What only it contributes. *That no parameter of this work is universal.* The closing grid is a staircase and the diagonal is the whole content: CODATA constants are universal, this work's numbers hold on a region or all elements and never universally. **A pooled fit across regions asserts a universal parameter, which the closed index says does not exist — that is a structural prohibition, not a stylistic preference, and it is what `domain_protocol.py` enforces.**

Rebuilt on three rulings: `kind` is the same axis as `source`; `arity` is a property of the book, not the parameter; and parameters that are artefacts of the METHOD are not physics of real atoms.

Λ_{amp} — the electron–electron term

What it is. The Slater integrals: for each subshell pair, F^k for k even up to $2\min(\ell, \ell')$ and G^k for $k \equiv \ell + \ell' \pmod{2}$ from $|\ell - \ell'|$ to $\ell + \ell'$. **Twenty cells, $E = 0$, a perfect triangle $0 \leq i \leq \ell$.**

What only it contributes. *That the rank is not a free coordinate.* Indexed on the multipole rank k the index gives $E = 1$ and the single defect is F^1 — **a term parity forbids.** Reindexed on POSITION WITHIN SEQUENCE it closes. **The same fault as `breadth` in Λ_{chem} : an axis whose values are constrained by another axis.**

And its structure explains the Slater result exactly: at s/s the exchange G^0 IS the direct F^0 , so dropping exchange costs nothing; at f/f it discards four independent quantities. **The count of lost integrals is the count of failures — 5 of 5 s -block brackets threaded, 0 of 1 f .**

Λ_{PCA} — physics $\oplus$ charge $\oplus$ amplitude, merged

What it is. Nine cells on $(\text{source}, \text{domain})$ with the domain refined by the charge regime: **universal** < **all elements** < **low** < **neutral** < **hydrogenic** < **one species**. $E = 0$.

What only it contributes. *The per-atom calibration.* A parameter's domain reads as a SET OF ATOMS, so for any (Z, c) the index states which parameters apply to it. **And the closing order puts `low` — charge 2 — BEFORE the neutral, which the ladders independently confirm: $c = 2$ is the only charge with two-sided brackets, the narrowest domain and therefore**

the most particular.

A correction recorded here: merging Λ_{phys} and Λ_{charge} on identified axis NAMES raised E from 6 to 11 and I concluded they were not one object. That was against a malformed Λ_{phys} . **Merged on** `domain  $\equiv$  regime` **alone, after the rebuild, they close.**

Λ_{chem} — the chemical properties

What it is. Forty-two chemical properties of a species on (kind, seat, PCA dependency). **E = 0 on fourteen cells** over the subvalence and valence shells.

- **kind:** count · symmetry · size · energy · rate - **seat:** the nucleus · the core · the subvalence shell · the valence shell · **the aggregate** - **PCA:** which of physics, charge or amplitude the property supplies

What only it contributes. *That twelve of the forty-two are properties of MATTER IN BULK and not of an isolated atom* — density, melting point, hardness, crystal structure, conductivity, colour of the metal, smell, taste, metallic character, reactivity. **With eleven hand-picked properties the index could not say this; it appeared the moment every property was demanded.**

And its closure states PCA's own boundary. E climbs **0 $\rightarrow$ 3 $\rightarrow$ 8 $\rightarrow$ 11 $\rightarrow$ 19** as the core, the nucleus and the aggregate are added. **PCA's domain is the subvalence and valence shells, and the index declares it by degrading monotonically outside it** — as Seaton's ratio is valid at $p = 0$ and undefined beyond.

The last defect cell, found by exhaustive reordering. All 1,440 orderings give minimum E = 1 and none reaches zero: the cell is **symmetry $\times$ the valence shell $\times$ a charge role**. The fill is the ground TERM, which changes with charge — $N_e = 20$ gives $^1S_0 \rightarrow ^3D_1 \rightarrow ^3F_2 \rightarrow ^3F_2$ and $N_e = 38$ the identical sequence. **Every ladder table carried the term symbol; the configurations were recorded and the terms were not.**

Λ_{t} — the corridor position

What it is. Twelve cells on (ℓ, n) holding **$t = (a - L)/(U - L)$** , where a sits in its corridor. **E = 3.**

$\ell \backslash n$	2	3	4	5	6	7	limit
s	0.268	0.335	0.266	0.218	0.257	0.195	—
p	·	1.120	1.049	1.022	1.002	·	1.0000
d	·	·	1.819	1.433	·	·	1.7321

What only it contributes. *That the ℓ axis carries the LIMIT and the n axis the APPROACH.* t converges to $\sqrt{\ell(\ell+1)/2}$, the centrifugal term — 6p sits 0.2% above it — and the gap falls **0.120 $\rightarrow$ 0.049 $\rightarrow$ 0.022 $\rightarrow$ 0.002** along n.

And q is not a coordinate. With the Pauli fraction in the radicand, t barely moves across a subshell: 6p gives 1.0237, 1.0417, 1.0009, 1.0036, 0.9946, 0.9940 across all six occupancies.

Its three defect cells are 7p, 6d and 7d — real absences, each a subshell whose corridor is one-sided or whose ionisation energy is not held.

Λ_{ladder} — the ladders themselves

What it is. The walks, on (seat, kind, Zcross). **FOURTEEN ladders in three families — six species, six state, two X-ray — nine cells, E = 0. IT CLOSES.** Registers 1356, 1374, 1375, 1384, 1427.

Two relations, three ladders each — **$Z = N_e + c - 1$** on Λ , and **$A = Z + N$** on the nuclide chart. Only two of three are independent either side, so no ladder varies Z alone: forbidden by arithmetic, not by lack of data.

family	ladder	fixes	reaches	held
species	isoelectronic	N_e	valence	11 of 11
species	the walk	c	valence	106 of 106
species	ionisation	Z	subvalence	15 of 108
species	isotopic	Z, N_e , c	nucleus	1 rung traced
species	isotonic	N	nucleus	none — NEW

family	ladder	fixes	reaches	held
species	isobaric	A	nucleus	none — NEW
state	the Rydberg series	the species	valence	every δ held
state	the ℓ -ladder	species, n	valence	62 pairs
state	the term ladder	species, cfg	valence	66 pairs
state	the outer-j ladder	species, cfg	valence	P.jsplit
state	the parent-term	the species	core	12 of 15
state	the isomeric	Z, N, e^-	nucleus	none — NEW

What only it contributes. *That a coordinate individuating the cells is a KEY, not an axis.* At four ladders *fixes* was injective by construction — a ladder IS named by what it holds fixed — so one cell per row, and **E = 0 was FORCED: 2% of admissible arrangements could refuse.** At six, ionisation and isotopic both fix Z, and the zero becomes earnable. That is the dual of A.define, where a single-valued coordinate contributes no envelope; this is the other end of the same degeneracy.

IT CLOSES, and both blocking reasons are resolved. The E = 1 defect stood at (**subvalence, counting, across elements**) — Moseley's ladder, in a direction ionisation does not go. That defect is what demanded Λ_{xray} , and Λ_{xray} filled it: Moseley enters at subvalence as counting/across, the $K\alpha$ doublet as coupling/within. **Only the subvalence seat closes** — nucleus gives 1, core 2, valence 1 — and the seat is settled from OUTSIDE the closure rather than by it, since Λ_{PCA} excluded the core independently, on different coordinates, before this was computed. The physics agrees: a $K\alpha$ line is not a property of the 1s shell but a transition *between* shells, so the seat is where the transition spans, not where the hole sits. **Contingency: 90% of comparable nine-cell sets refuse, so the zero is earned** (R 1427).

And the five shared cells are four findings and one separation (R 1428). Two pairs are ONE OBJECT and the index is right to merge them — isotonic fixes N and isobaric fixes A with $A = Z + N$; isoelectronic fixes N_e and the walk fixes c with $Z = N_e + c - 1$. That is *charge is one symbol in three positions*, one level up. Two are separated by axes already in the index but not among the three that close: isotopic is species where the isomeric is state; the Rydberg series fixes the species where the ℓ -ladder fixes species and n. **The fifth was the real ambiguity** — the term ladder and the outer-j ladder — and they are five tower stages apart, nested rather than parallel: the term ladder moves 2S, Λ_8 's own letter, while the outer-j ladder moves 2J, which arrives only at Λ_{13} . Adding the tower stage as a fourth axis separates that pair and only that pair, but costs closure, E going $0 \rightarrow 1$. **So the stage is the discriminator, not an axis** (R 1429).

Λ_{xray} — the inner shell

What it is. The dipole-allowed inner-shell transitions, on $(\Delta n, \Delta \ell, j\text{type_hole})$. **33 lines through the N shell $\rightarrow$ 9 cells, E = 0.** Register 1378.

Where it comes from. Built with no new data. **Λ 's cell IS a transition** — source subshell into target subshell — and its constraints impose no order between n and e, so a downward transition was already inside its alphabet; only the caps excluded it. EM.map gives the dipole rule, T.a12 the j triangle. The six K-shell lines emerge as the classical set unadjusted: K-L2 and K-L3 are $K\alpha_2$ and $K\alpha_1$, K-M2 and K-M3 are $K\beta_3$ and $K\beta_1$, K-N2 and K-N3 are $K\beta_2$. Energies are NIST SRD 128, Deslattes *et al.*, *Rev. Mod. Phys.* **75** (2003) 35–99.

What only it contributes. *That thirty-three named lines are nine transition types.* Seventy-two of 165 three-coordinate systems reach E = 0 and **every one collapses to exactly nine cells** — $K\alpha$, $L\alpha$ and $M\alpha$ are one type read at three depths, which Siegbahn notation hides.

And it settles what Λ_{cross} 's silences mean. The two close on the same shape — differences plus one endpoint — and the cypher separates them:

language	Λ_{cross}	Λ_{xray}
order	speaks	speaks, E = 0
geometry	speaks	speaks
statistics	silent	speaks — pairwise marginals recover all 9
analysis	silent	speaks — Moseley, $R^2 = 0.998$

Λ_{cross} remains the only integer object: cells AND values fixed by arithmetic. Λ_{xray} has integer cells and measured values. Registers 1379, 1380, 1385.

The one-line summary, extended

index	the one thing only it gives
Λ_{law}	the carrier — which variable a law runs in
Λ_{const}	that <i>standing</i> cannot be a coordinate
Λ_{var}	no nucleus + rydberg cell — the three-body factorisation
Λ_{ryd}	the n-dependence Λ_{spectra} discards
Λ_{charge}	c is three coordinates in one symbol
Λ_{cross}	an index with no statistics language
Λ_{descent}	the only place a fit belongs
Λ_{phys}	no parameter of this work is universal
Λ_{amp}	the rank is not a free coordinate
Λ_{PCA}	the per-atom calibration
Λ_{chem}	twelve of forty-two properties are not an atom's at all
Λ_{t}	ℓ carries the limit, n carries the approach
Λ_{ladder}	that a coordinate individuating the cells is a key, not an axis
Λ_{xray}	thirty-three named lines are nine transition types

Fourteen close. Λ_{xray} is new and closes at nine transition types. **Λ_{ladder} now closes too** — fourteen ladders, nine cells, $E = 0$, once the X-ray pair is seated at subvalence (R 1427), its five shared cells resolving as four findings and one separation (R 1428–1429).

Λ_{spectra} closes at the LIMIT — not for want of the principal number, as register 1341 supposed. The index runs to the last available species and stops: 98 cells, $E = 58$, of which 38 need more electrons than any atom has. **The remaining twenty are within the limit, and eleven of those are the Madelung exceptions, each nameable** (R 1395, corrected at R 1426).